



LDO Regulator

Project Name: Digital High Frequency Chip – Linear Low Dropout Voltage Regulator Project Number: 3097

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Project carried out at: University

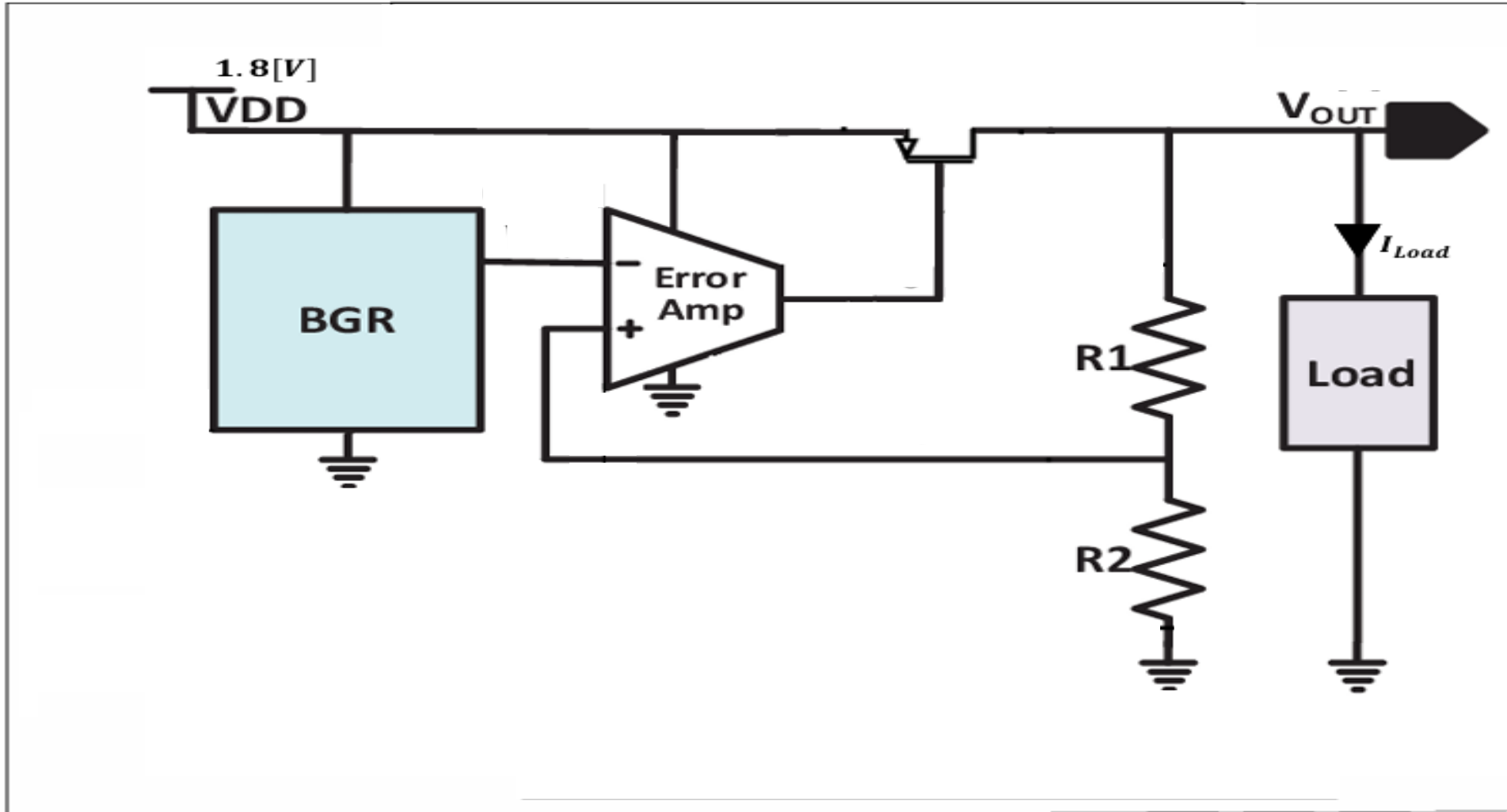
Instructor's Signature: Tal Elazar



Abstract and Introduction

- The project's objective is to design an LDO for a DPLL.
- DPLLs require a highly stable voltage input for proper operation.
- Variations in **supply voltage**, **temperature**, and **load current** can degrade performance.
- The LDO is necessary to maintain a constant input voltage despite these variations.
- A well-designed bandgap reference circuit and a high-gain stable OTA are essential for the LDO's design.

LDO Operating Principle





Engineering Motivation

■ What is a DPLL?

- A **Digital Phase-Locked Loop (DPLL)** is a closed-loop system that synchronizes an internal oscillator to a reference clock.
- It is commonly used in **clock generation, frequency synthesis, and data recovery** in modern SoCs.
- The reference clock is typically generated by a **quartz crystal oscillator**, which provides a **precise but low-frequency signal**.
- The DPLL uses this reference to **generate a higher-frequency, low-jitter clock** by continuously adjusting its internal oscillator (VCO) through feedback.



Engineering Motivation

Why Clean Supply Matters: VCO & TDC

- **VCO (Voltage-Controlled Oscillator)** generates a clock signal whose **frequency depends on voltage**.
 - Supply noise causes **jitter** and frequency errors.
 - **TDC (Time-to-Digital Converter)** measures time differences with **picosecond accuracy**.
 - Voltage fluctuations lead to **timing inaccuracy**.
-  Both blocks are **highly sensitive** to power supply variations.
-  A stable and low-noise LDO is essential to maintain **DPLL performance**.

Target Specifications

To ensure proper operation of the DPLL, the LDO must meet the following key requirements:

- ◆ **Line Regulation** - Measures how much the output voltage changes with variations in input voltage. $\left(\frac{\Delta V_{out}}{\Delta V_{DD}}\right)$ **Target:** $\leq 10 m \left[\frac{V}{V}\right]$ across VDD variations
- ◆ **Load Regulation** - Indicates stability of the output under changing load currents. $\left(\frac{\Delta V_{out}}{\Delta I_{load}}\right)$ **Target:** $\leq 100 m \left[\frac{V}{A}\right]$ from 0 to 60mA.
- ◆ **Temperature Coefficient** - Measures sensitivity of the output voltage to temperature changes. $\left(\frac{\Delta V_{out}}{\Delta Temp \cdot V_{out}} \cdot 10^6\right)$ **Target:** $\leq 60 ppm/^{\circ}C$ (parts per million per degree)



Design Approach – Meeting the Spec

To meet the key performance requirements, the following design strategies were applied:

◆ Line Regulation/PSRR

→ Use a **high-gain error amplifier** to maintain a stable output under VDD variations

◆ Load Regulation

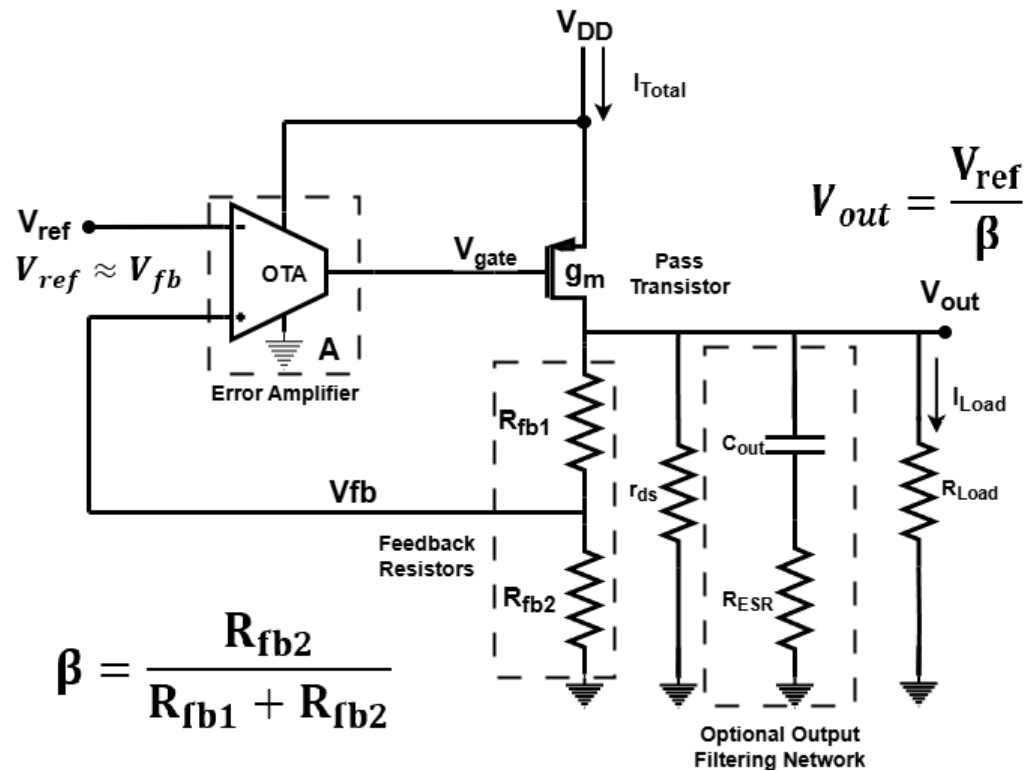
→ Design a **strong pass transistor** capable of sourcing the maximum load current

◆ Temperature Coefficient

→ Integrate a **precise Bandgap Reference (BGR)**

→ The BGR includes its own **high-gain error amplifier** to ensure thermal stability

Required OTA Gain



◆ $LR = \frac{\Delta V_{out}}{\Delta V_{DD}} \approx \frac{1}{\beta A} \rightarrow A_{OTA} > 60dB \text{ for } 1m \frac{V}{V}$

◆ $LDR = \frac{\Delta V_{out}}{\Delta I_{Load}} \approx \frac{1}{\beta g_m A} \rightarrow A_{OTA} > 60dB \text{ for } 10m \frac{V}{A}$

◆ Simplified optimistic small-signal model.

⚠ **In practice**, 6-8dB more gain are needed to make up for layout and parasitic effects.



Full Project Requirements

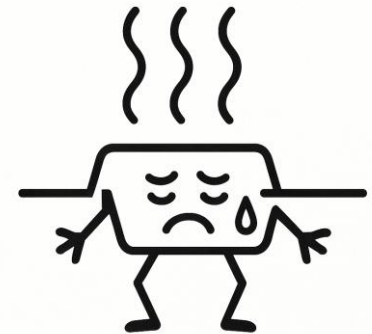
Specification	Definition	Required Value
BGR Output Voltage	V_{ref} [V]	0.6–0.8V
LDO Output Voltage	$V_{\text{out}} = V_{\text{ref}} \left(1 + \frac{R_{\text{fb1}}}{R_{\text{fb2}}} \right)$ [V]	0.9V
LDO Max Load Current	$I_{\text{LOAD,max}}$ [mA]	55.5mA
Line Regulation	$\frac{\Delta V_{\text{OUT}}}{\Delta V_{\text{DD}}}$ [V/V]	< 0.01
Load Regulation	$\frac{\Delta V_{\text{OUT}}}{\Delta I_{\text{LOAD}}}$ [mV/mA]	< 0.1 from 0–60mA
Temperature Coefficient	$\frac{V_{\text{OUT,max}} - V_{\text{OUT,min}}}{V_{\text{OUT,avg}} \times \Delta T} \times 10^6$ $\left[\frac{\text{ppm}}{^\circ\text{C}} \right]$	< 60 from -40° to 125°
Power Supply Rejection Ratio (PSRR)	$20 \log_{10} \left(\frac{\Delta V_{\text{OUT}}}{\Delta V_{\text{DD}}} \right)$ [dB]	< –40dB
Quiescent Current	$I_Q = I_{\text{Total}} - I_{\text{Load}}$ [mA]	< 1mA
OTA Performance	A_v [dB], GBW [Hz], Phase Margin $[\circ]$	> 70 dB, > 5 MHz, > 60°

Why Temperature Matters in Analog Circuits

Temperature has a strong influence on key electrical components:

- **Resistors** – resistance increases with temperature
- **Diodes and BJTs** – V_{BE} decreases with temperature
- **MOSFETs** – mobility, threshold voltage, and leakage are temperature-dependent
- **Thermal noise** increases as temperature rises

The **Bandgap Reference (BGR)** is a circuit that generates a voltage reference that remains stable over temperature.





BGR – Operating Principle

Key Principle – Simple but Brilliant:

Combine two voltages with opposite temperature behaviors:

One that decreases with temperature (like V_{BE})

One that increases with temperature (like a PTAT voltage)

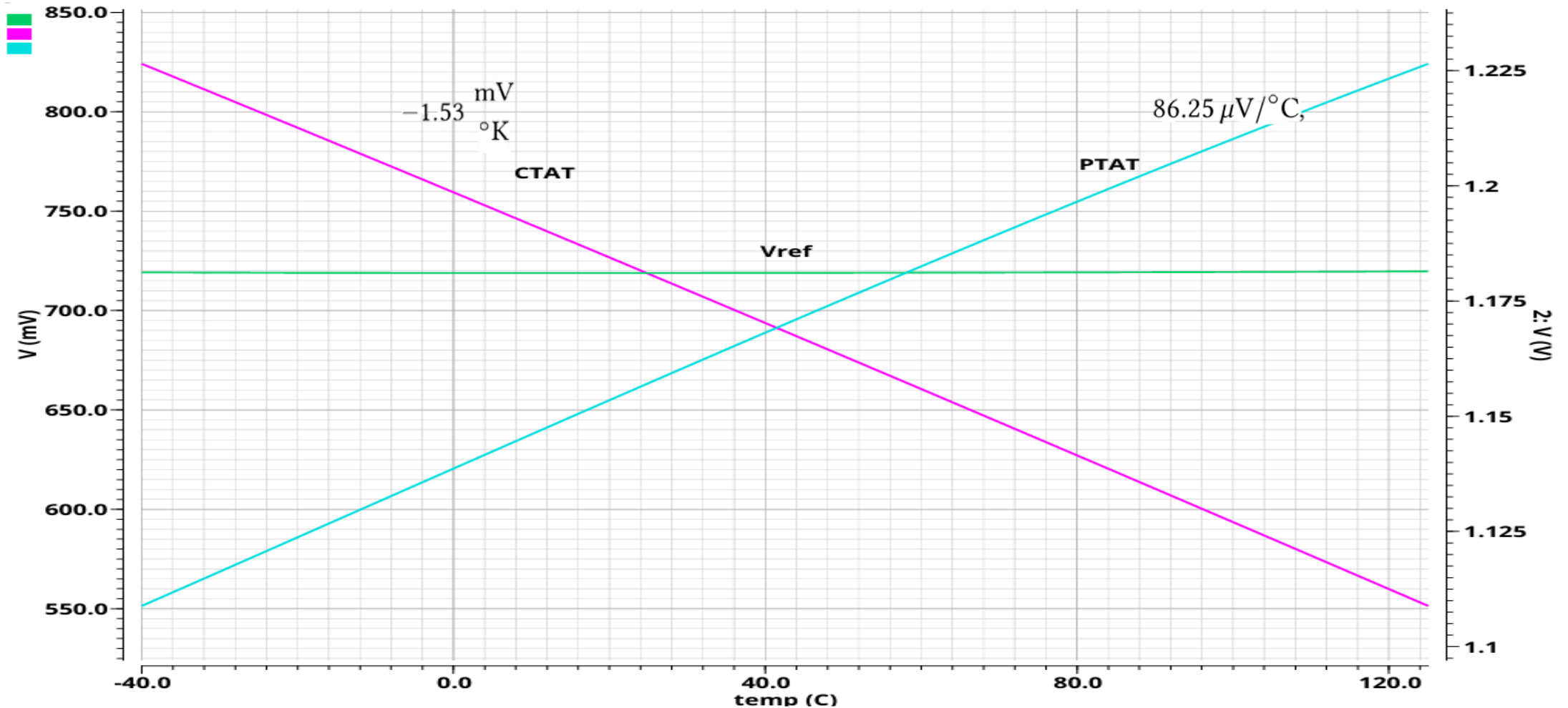
→ By adjusting their weights and summing them, the temperature dependence is canceled out.

(*) In most designs, the output of a BGR is around 1.2 V.

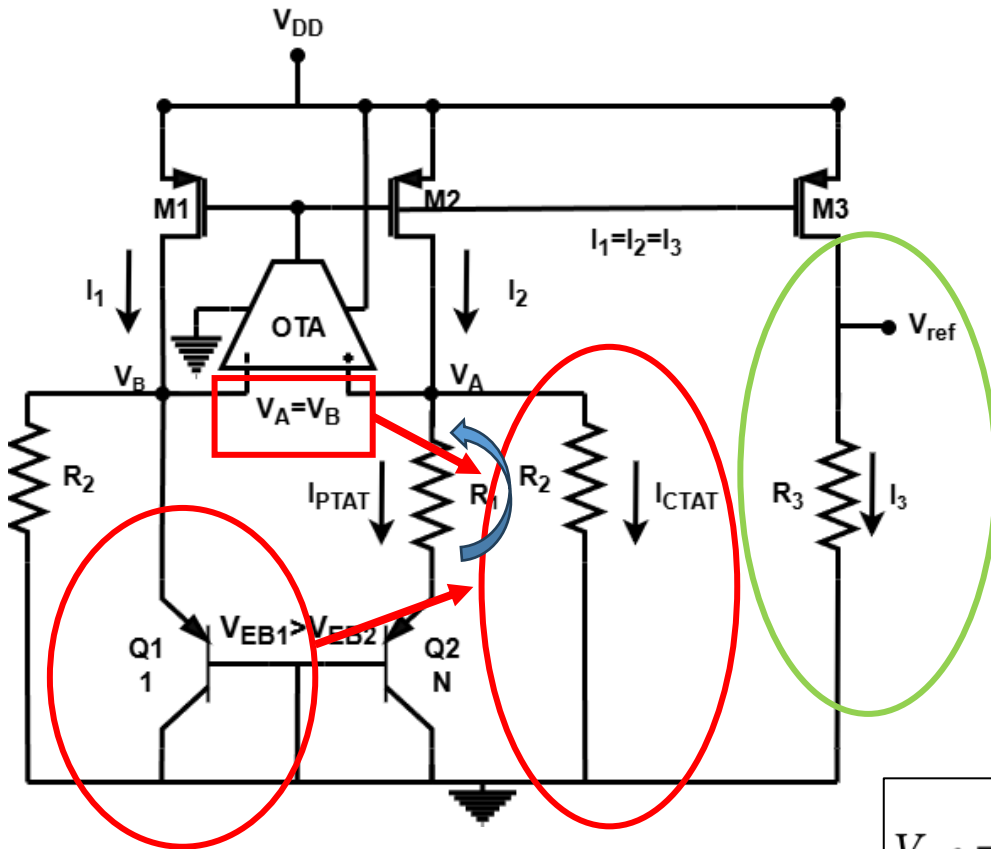
In our implementation, we required a **lower reference voltage** to meet the LDO output target. Therefore, we used a **slightly different current-mode topology**, but the core temperature compensation principle remains the same.

BGR – Operating Principle

Demonstration Of BGR



BGR Operating Principle



$$V_{EB} = V_T \ln \left(\frac{I_C}{I_S} \right) , V_T = \frac{kT}{q}$$

$$I_{CTAT} = \frac{V_{EB}}{R_2} \text{ (CTAT current)}$$

$$\Rightarrow \left. \frac{dV_{EB}}{dT} \right|_{T=300\text{ K}} = -1.53 \frac{\text{mV}}{^\circ\text{K}}$$

$$I_{\text{PTAT}} = \frac{V_{\text{EB}_1} - V_{\text{EB}_2}}{R_1} = \frac{kT}{qR_1} \ln(N) \quad (\text{PTAT current})$$

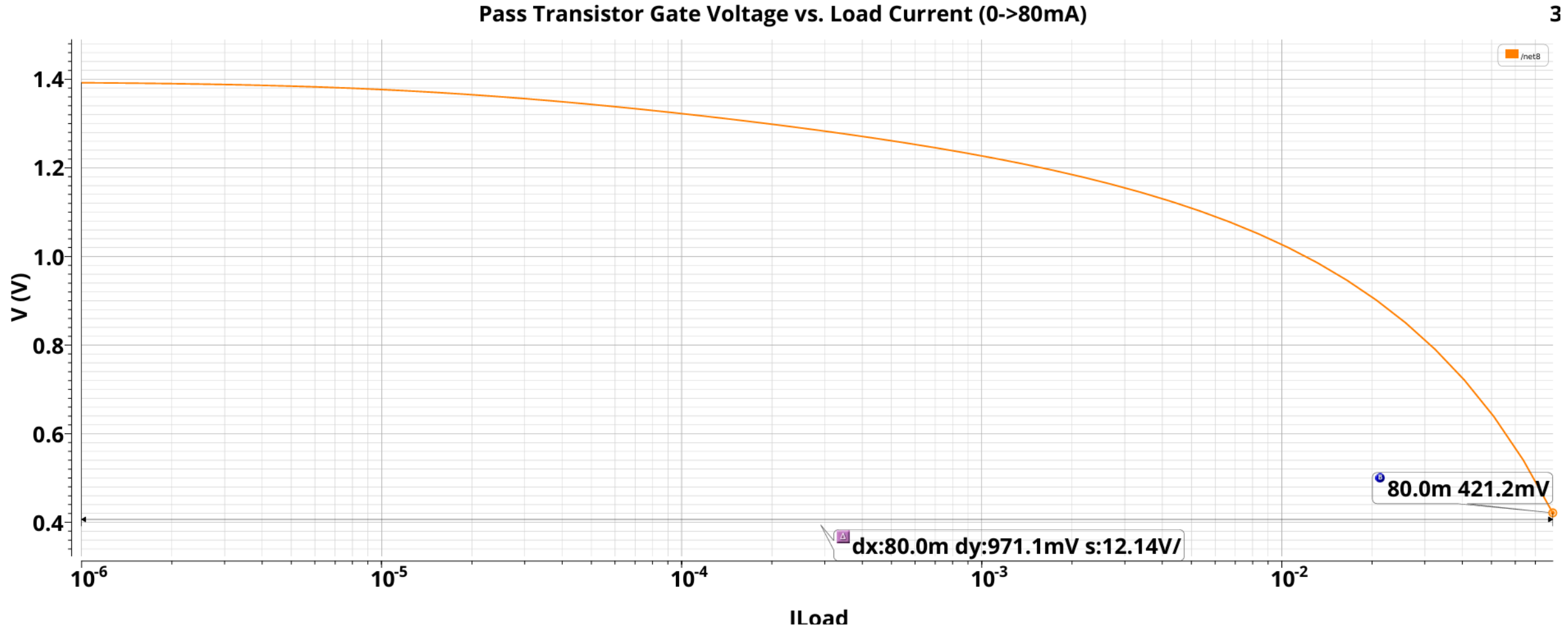
$$\Rightarrow \frac{dV_T}{dT} = \frac{k}{q} \approx 86.25 \mu\text{V}/^\circ\text{C},$$

$$V_{\text{ref}} = R_3 (I_{\text{PTAT}} + I_{\text{CTAT}}) = R_3 \left(\frac{kT}{qR_1} \ln(N) + \frac{V_{\text{EB}}}{R_2} \right) = R_3 \underbrace{\left(\frac{\ln(N)}{R_1} V_T + \frac{V_{\text{EB}}}{R_2} \right)}$$

Temperature independent if TCs cancel

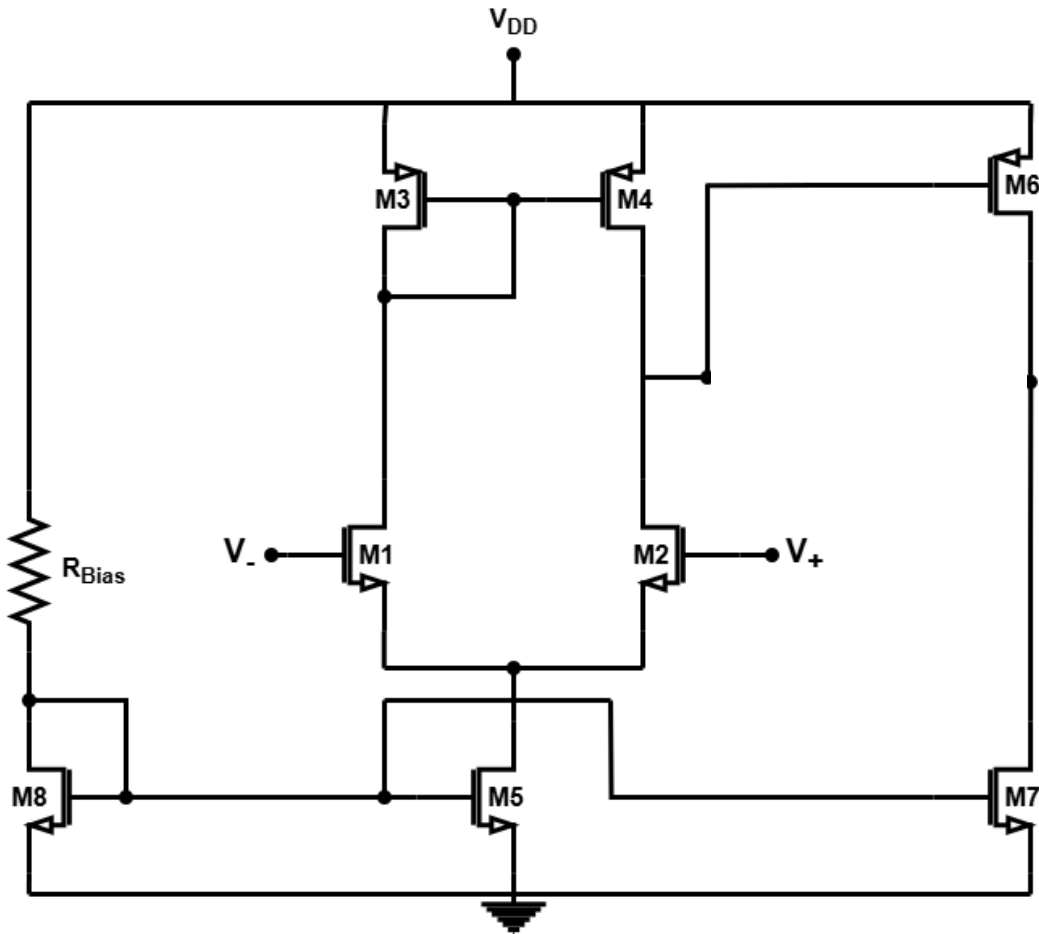


The Role of the OTA in Our Design



3

Two-Stage Miller OTA



High Gain

$$A_{v1} = g_{m2} \cdot (r_{ds2} \parallel r_{ds4}) , A_{v2} = g_{m6} \cdot (r_{ds6} \parallel r_{ds7})$$

$$\Rightarrow A_{tot} = g_{m2} \cdot (r_{ds2} \parallel r_{ds4}) \cdot g_{m6} \cdot (r_{ds6} \parallel r_{ds7})$$

$$(*) \text{ for } A_{v1} \approx A_{v2} \triangleq A_{CS} , A_{tot} = (A_{CS})^2$$

Frequency compensation and stability

$$p_1 \approx \frac{1}{R_{out1} A_{tot} \cdot C_c} \Rightarrow GBW = A_{DC} \times p_1 = \frac{g_{m1}}{C_c}$$

$$p_2 \approx \frac{g_{m6}}{C_L}$$

$$z \approx \frac{1}{C_c \left(\frac{1}{g_{m6}} - R_c \right)} \text{ LHP Zero for: } R_c > \frac{1}{g_{m6}}$$



From Schematic to Layout

Once the schematic design was finalized and verified through simulations, the circuit was implemented in physical layout using **TSMC 28nm CMOS technology**.

Layout quality directly affects:

- ✓ Performance (gain, matching, parasitics)
- ✓ Area and density
- ✓ Noise and stability

Special attention was given to:

Symmetry in analog blocks (e.g., OTA differential pair)

Matching techniques for current mirrors and resistors

Routing of sensitive nodes (e.g., BGR core, OTA outputs)

Minimization of IR drop through metal stacking

Layout – It's All About Matching

In analog layout, **perfect interconnects don't exist** —
Parasitic **capacitances&resistances** will always be there.
The key idea behind **matching** is not to eliminate these effects,
but to ensure they impact all matched devices **equally**.

Techniques we applied:

- ✓ **Common-centroid** layout for critical pairs
- ✓ **Dummy devices** to minimize edge mismatches
- ✓ **Guard rings** around sensitive transistor clusters to isolate substrate noise
- ✓ **Symmetric routing** to balance parasitics

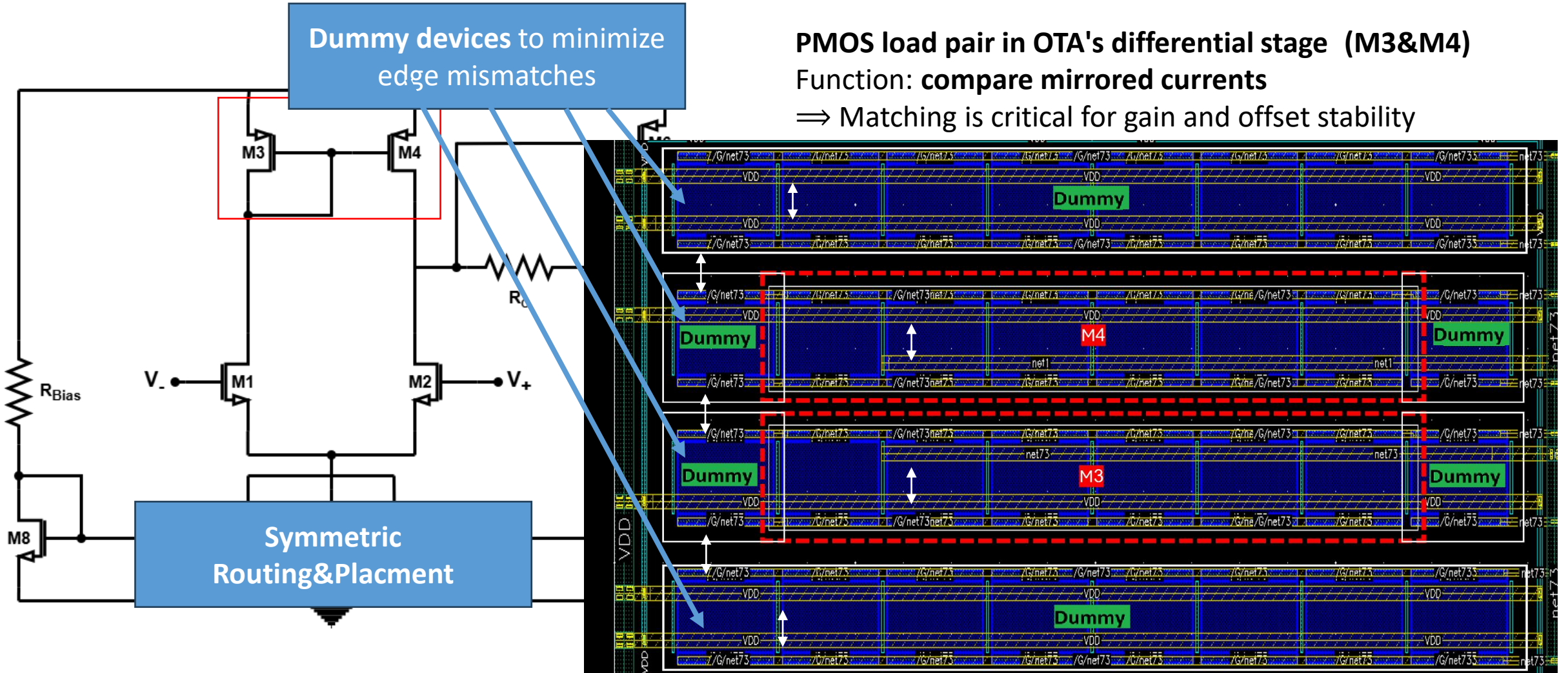
For Example 1

Dummy devices to minimize
edge mismatches

PMOS load pair in OTA's differential stage (M3&M4)

Function: **compare mirrored currents**

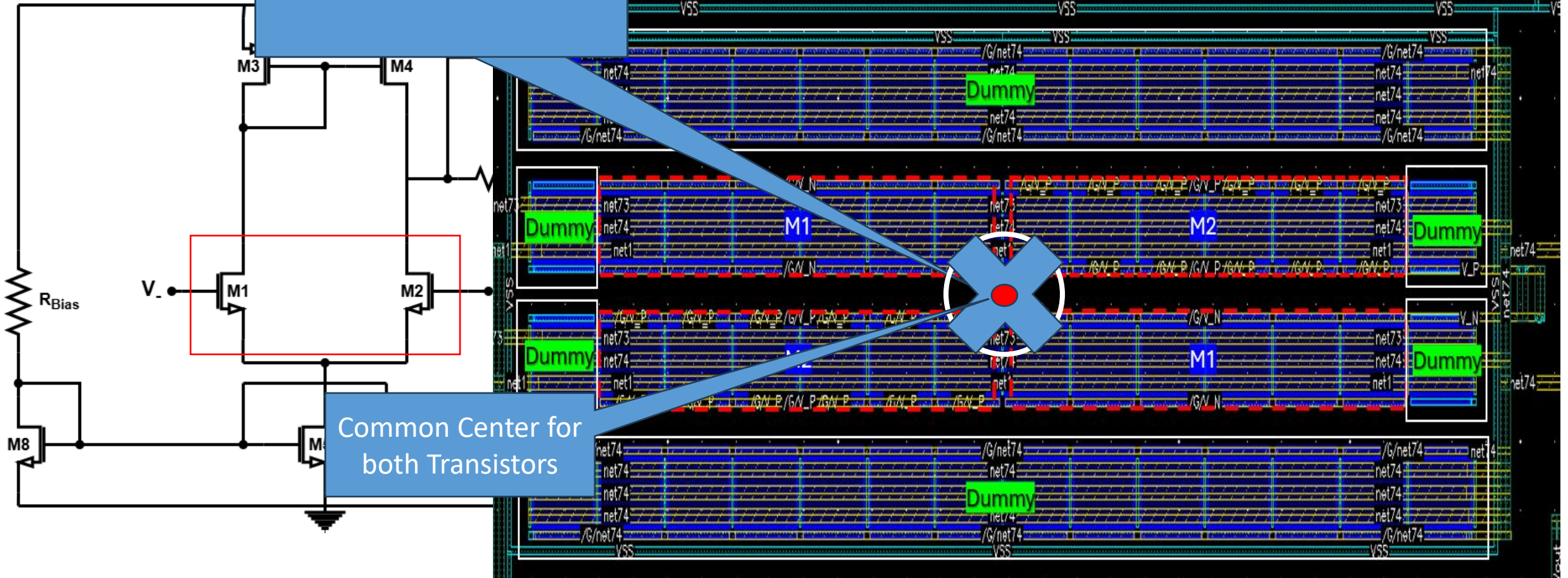
⇒ Matching is critical for gain and offset stability



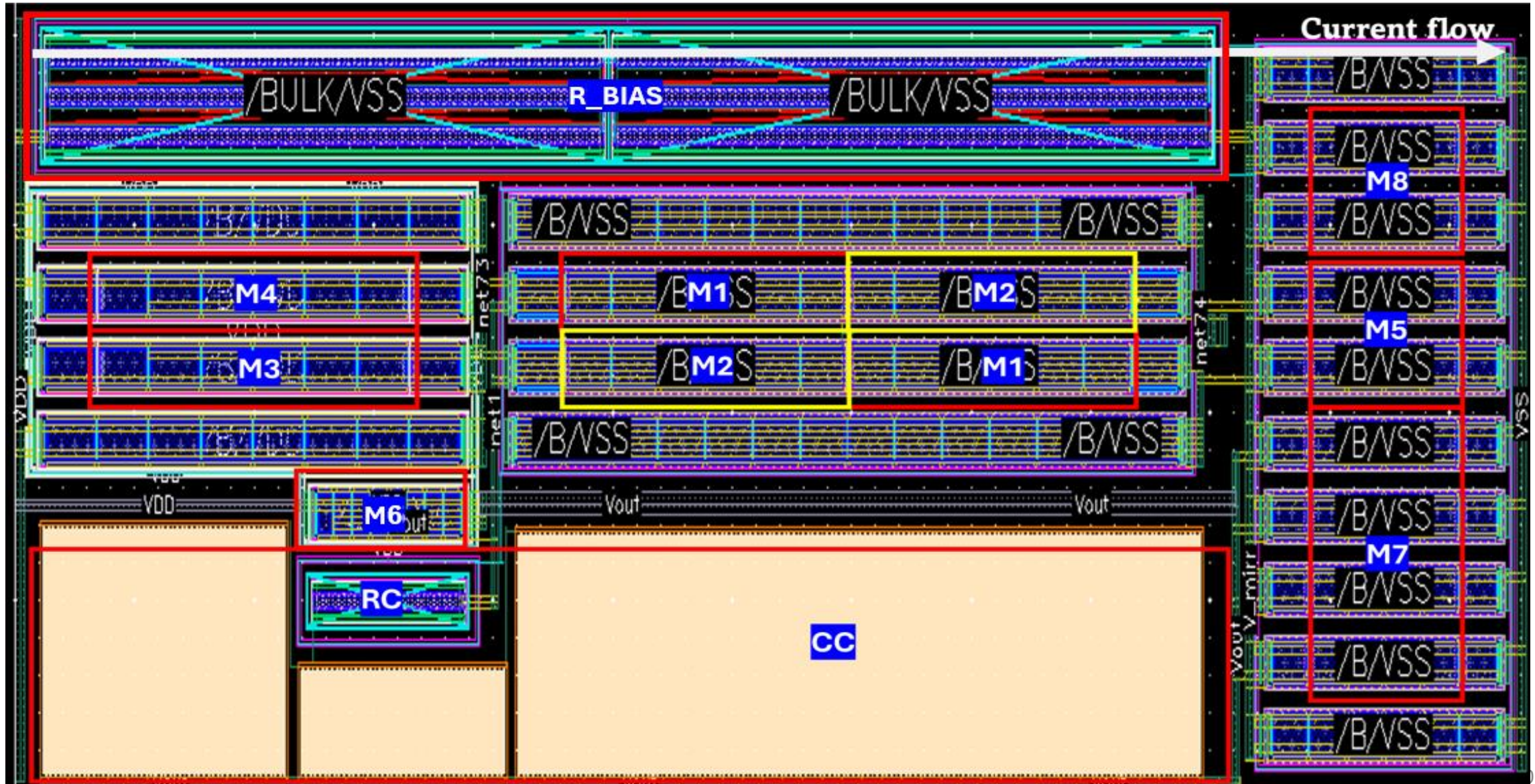
For Example 2

High Symmetry routing
between the devices

Differential Pair (M1 & M2)

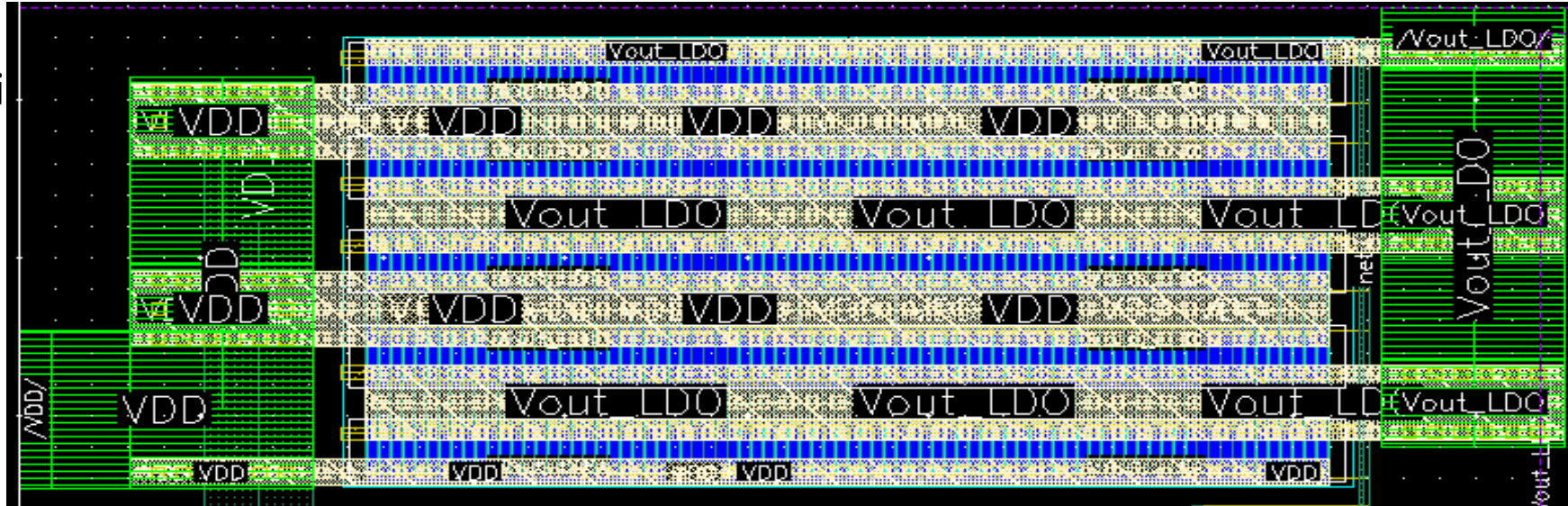


Final OTA Layout



A Bit About IR Drop

IR drop i

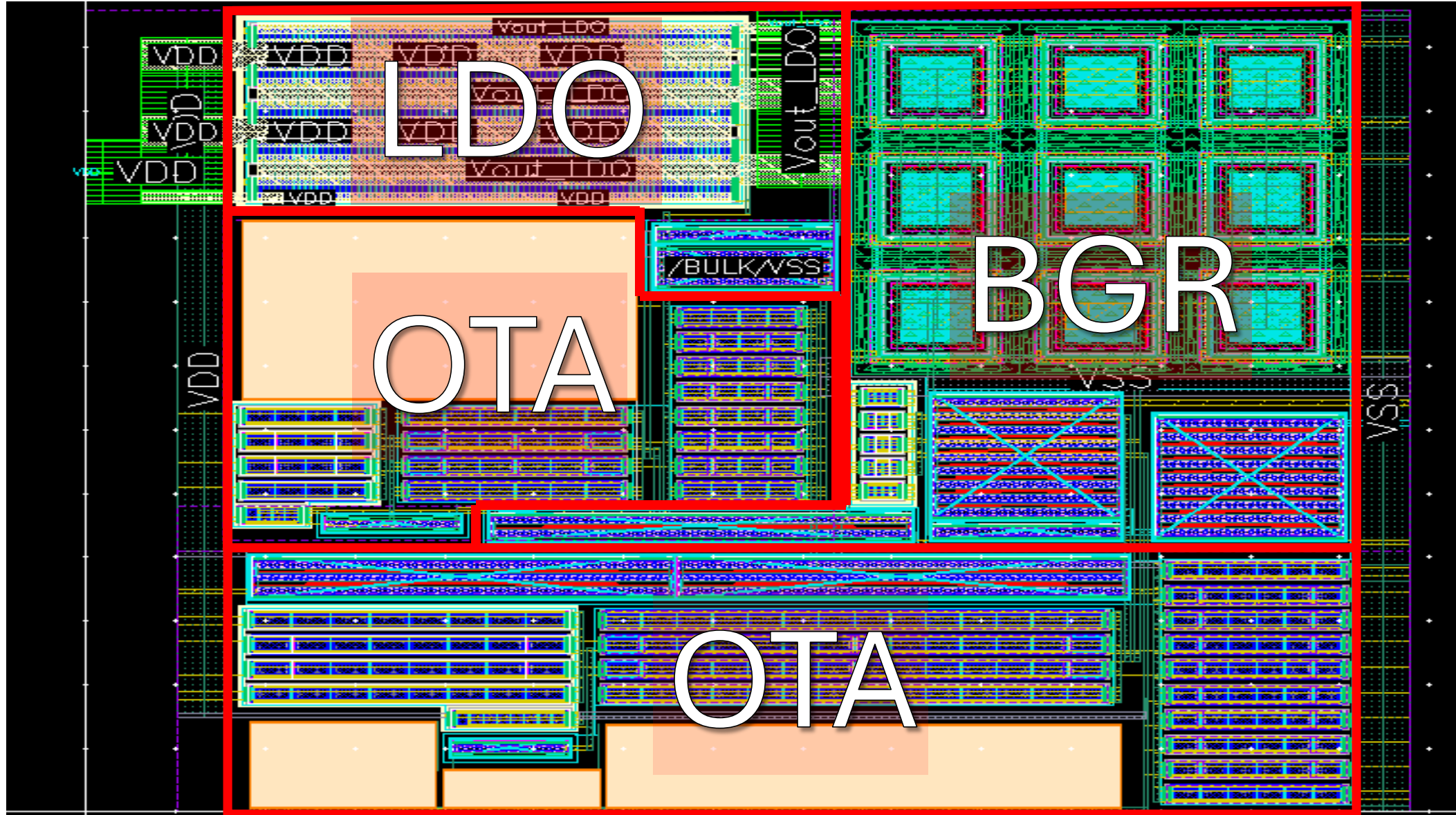


This affects the **actual voltage delivered** to the load, especially in analog blocks.

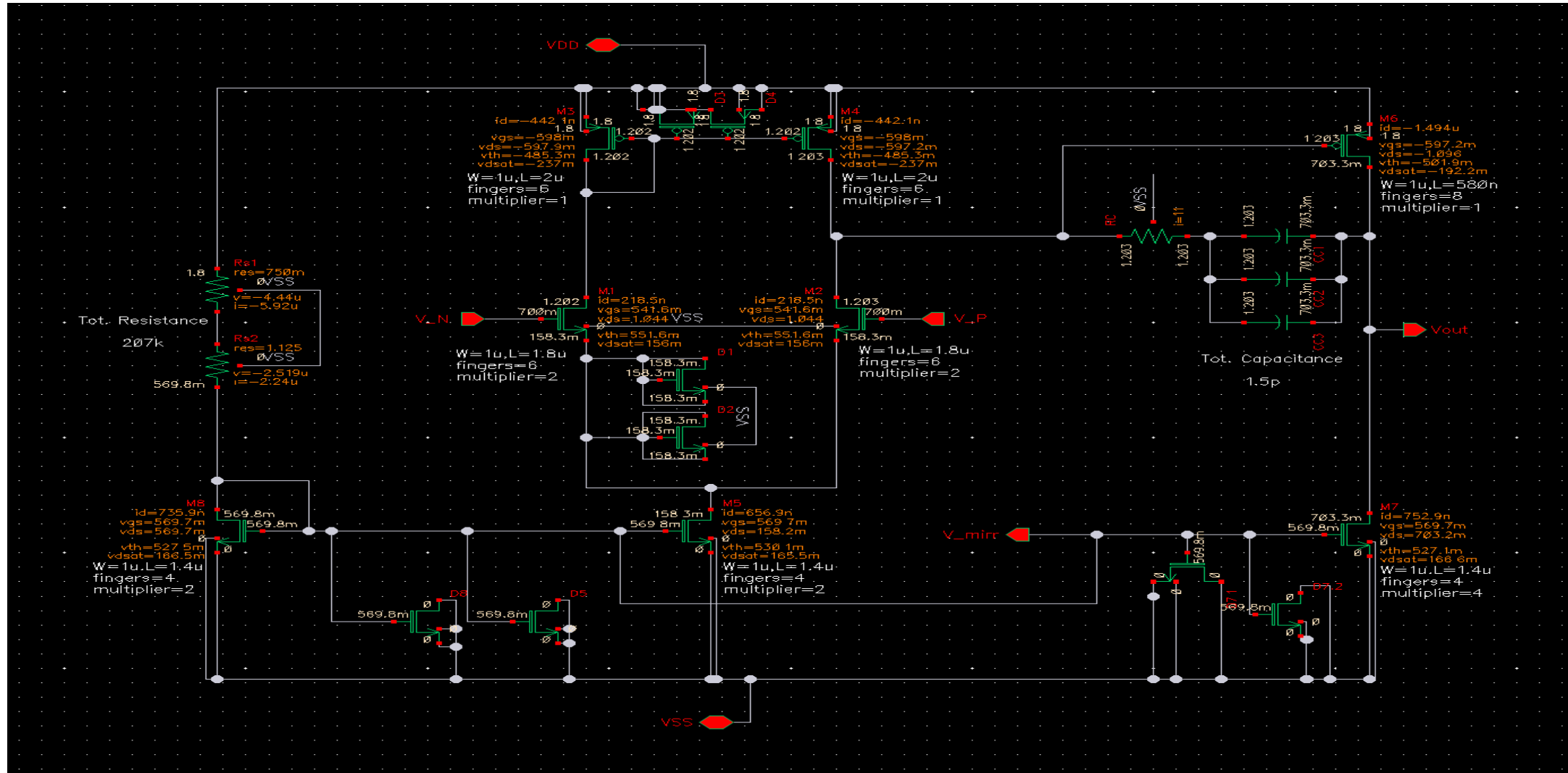
To ensure proper LDO operation at **60 mA load**, careful layout planning was required:

-  **Wire width and spacing** calculated according to PDK to pass DRC
-  **Multiple metal layers** selected based on current density
-  **Metal stacking** used to distribute current across layers to further minimize resistance
-  **Segmented connections** made to match physical transistor layout

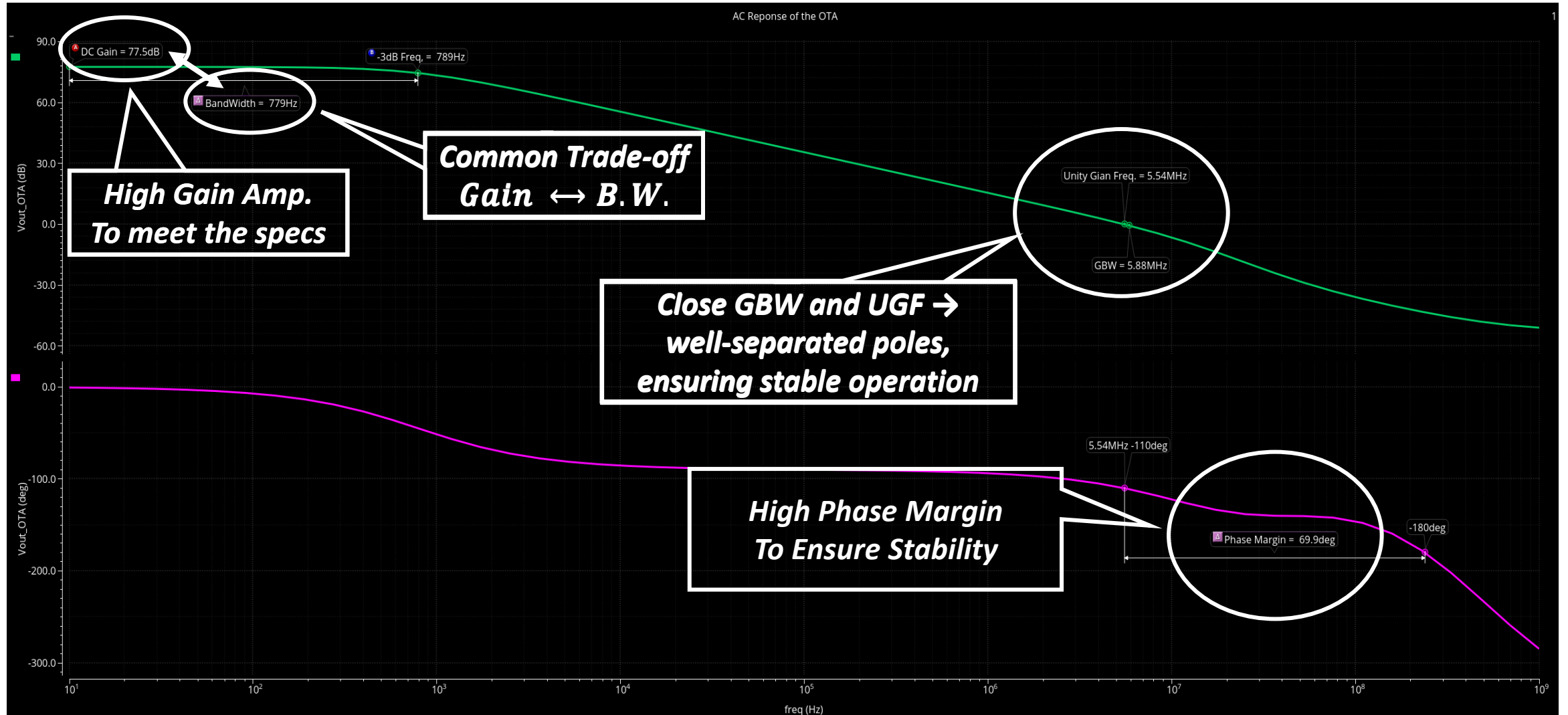
Final Layout



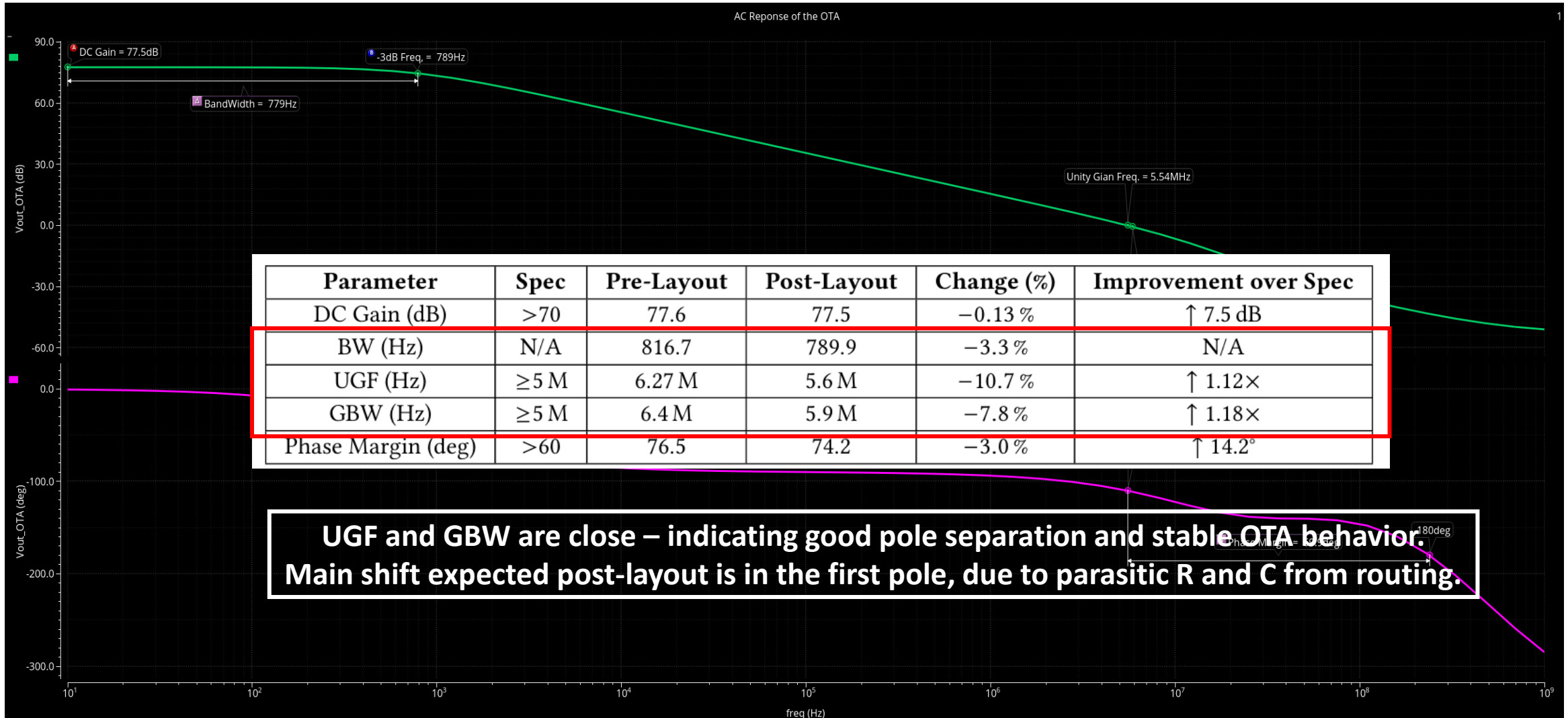
OTA - Schematic



OTA – Frequency Response (Post Layout)

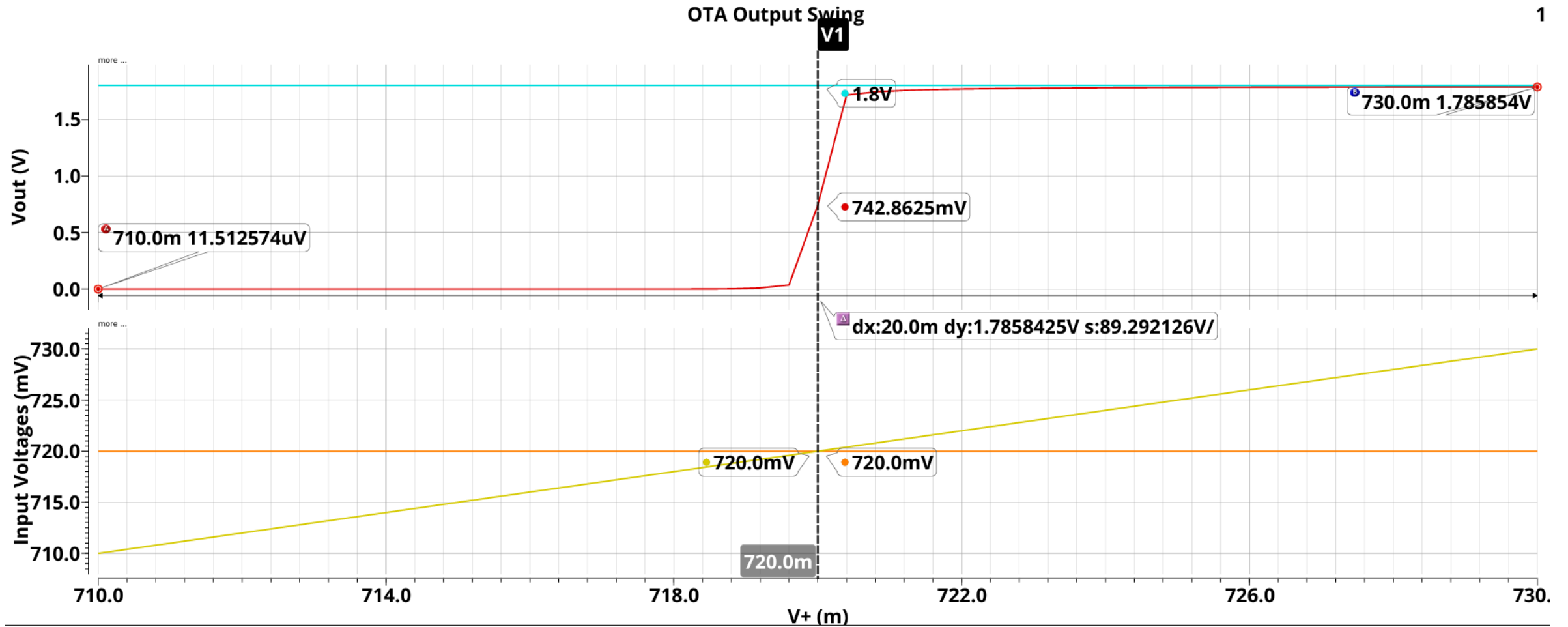


OTA – Frequency Response (Post Layout)

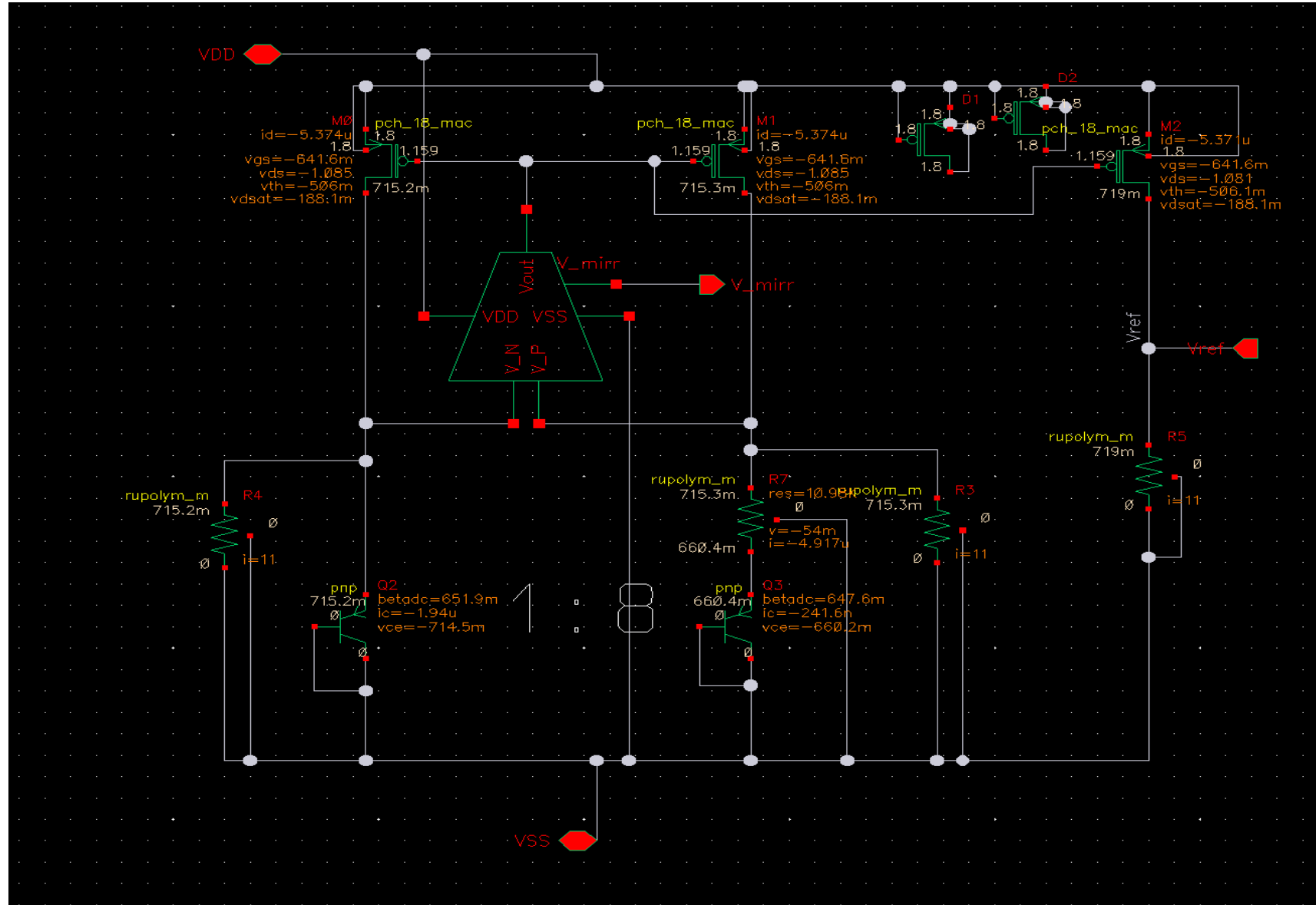




OTA – Output Swing

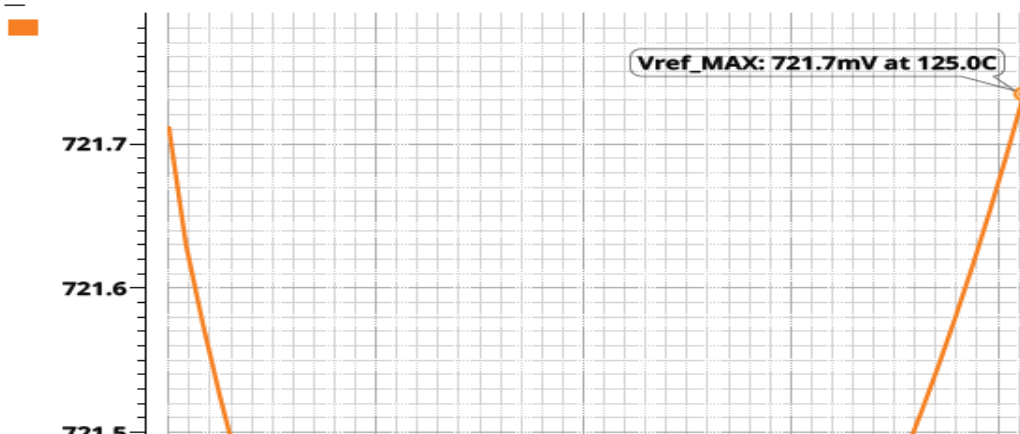


BGR - Schematic

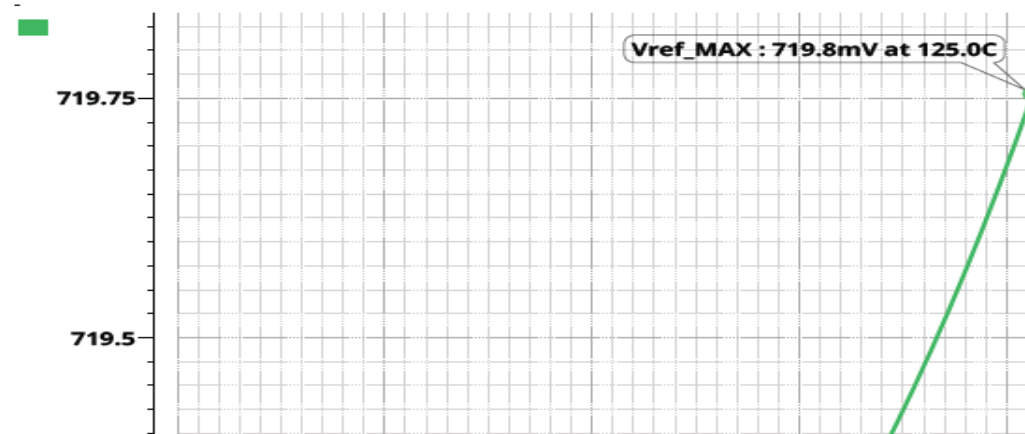


BGR – Temperature Stability

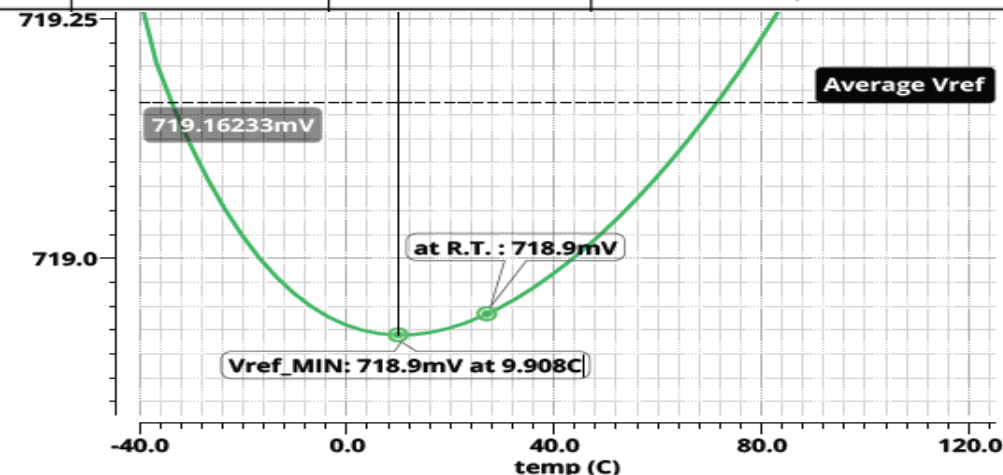
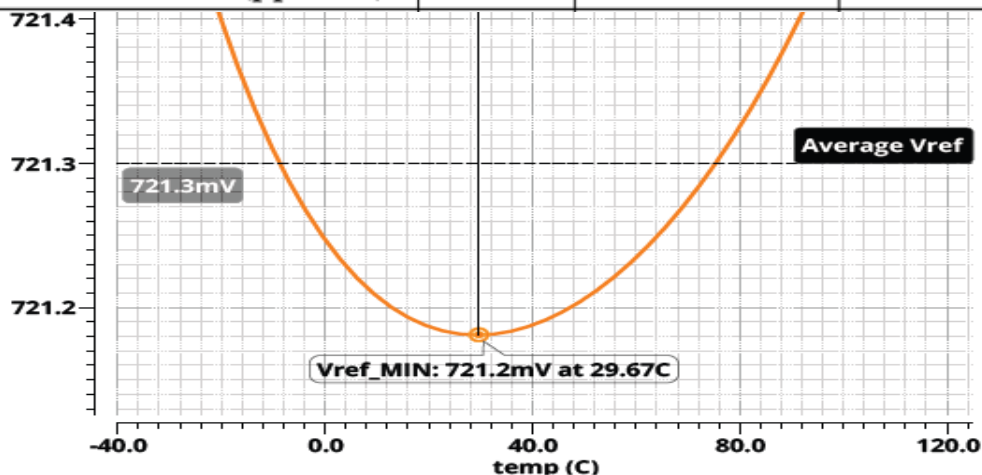
Temperature Coefficient - BGR (Pre LayOut)



Temperature Coefficient - BGR (Post LayOut)



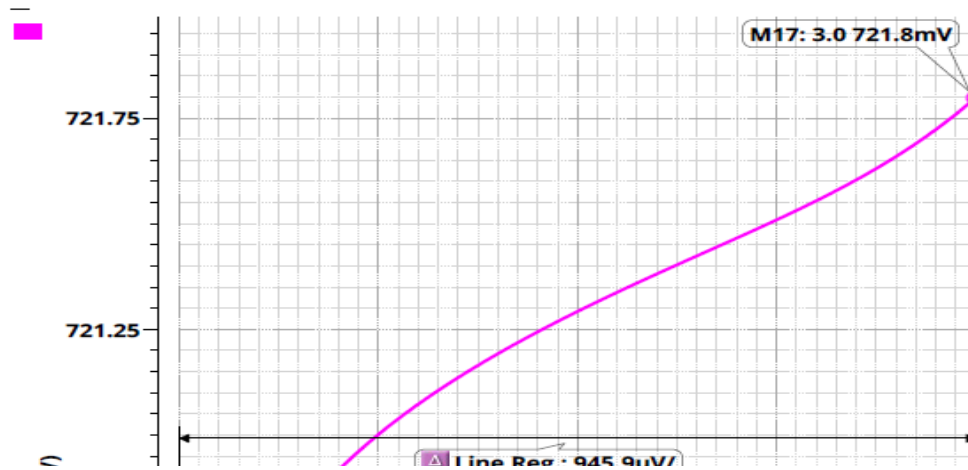
Parameter	Spec	Pre-Layout	Post-Layout	Range	Change (%)	Improvement Over Spec
Temperature Coefficient (ppm/°C)	<60	4.6	7.03	-40-125 °C	+52.8 %	↑ 8.5×



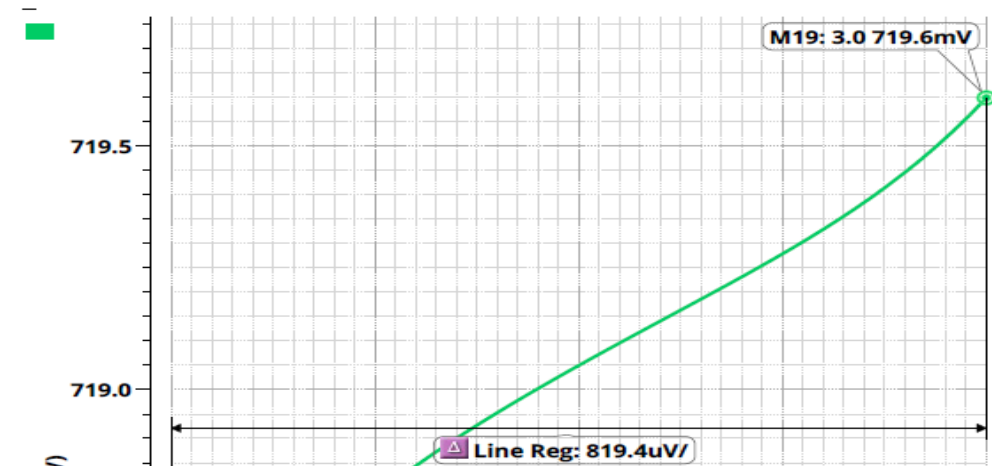
Temp Sweep : -40 - 125 [C]

BGR – Line Regulation

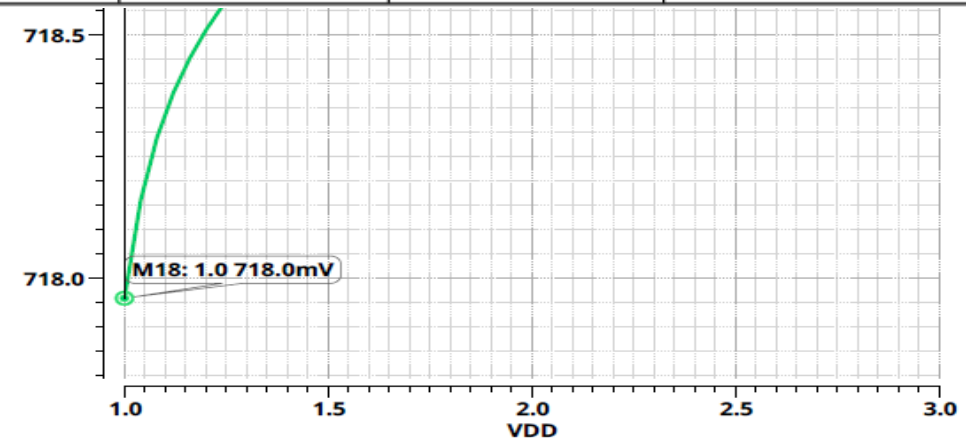
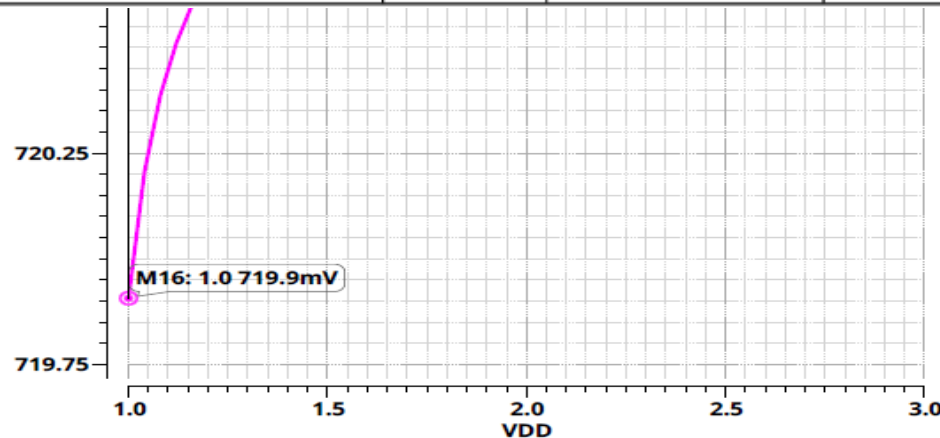
Line Regularization - BGR (Pre LayOut)



Line Regularization - BGR (Post LayOut)

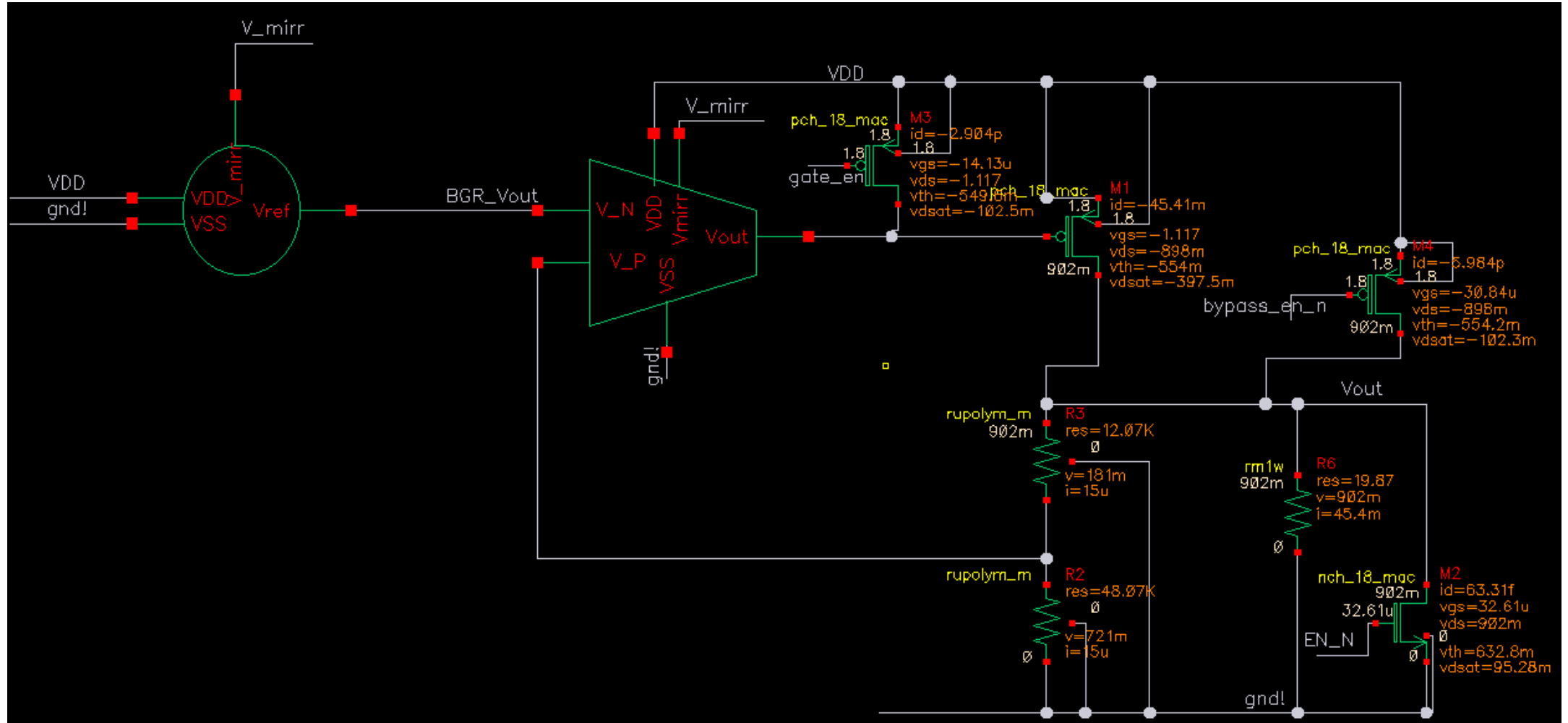


Parameter	Spec	Pre-Layout	Post-Layout	Range	Change (%)	Improvement Over Spec
Line Regulation (V/V)	<0.01	0.00058	0.00054	1.4–2.2 V	–6.9 %	↑ 18×



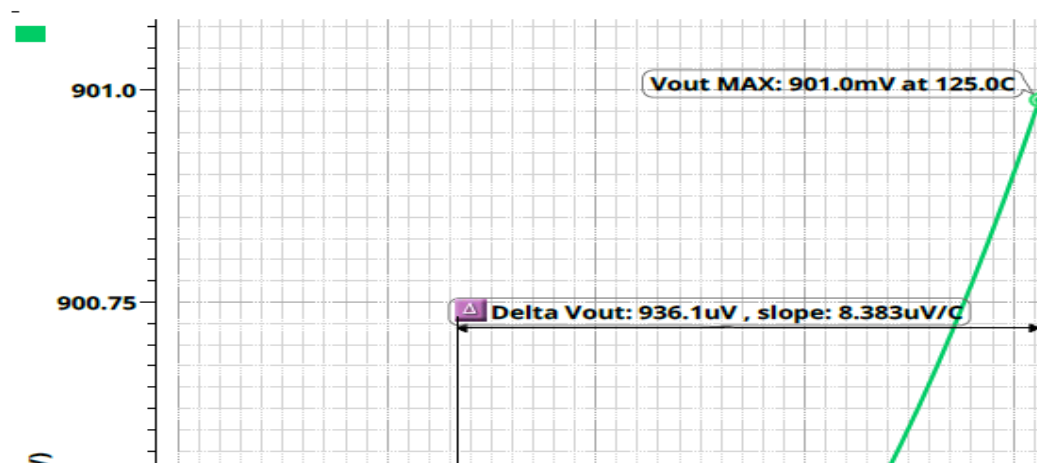
VDD Sweep : 1-3 [V]

LDO - Schematic

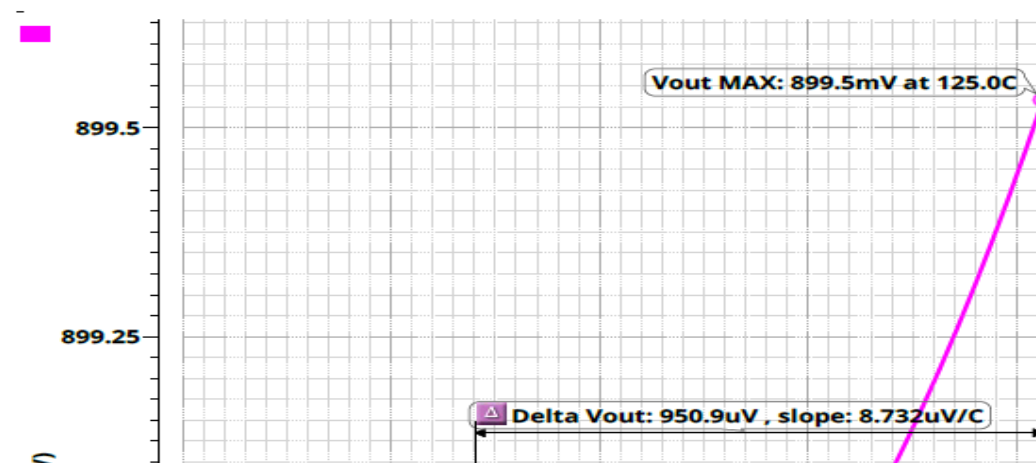


LDO – Temperature Stability

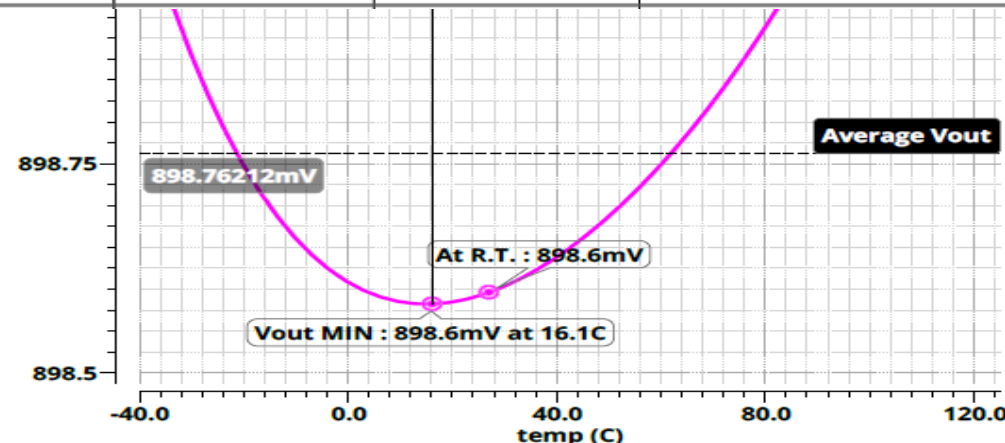
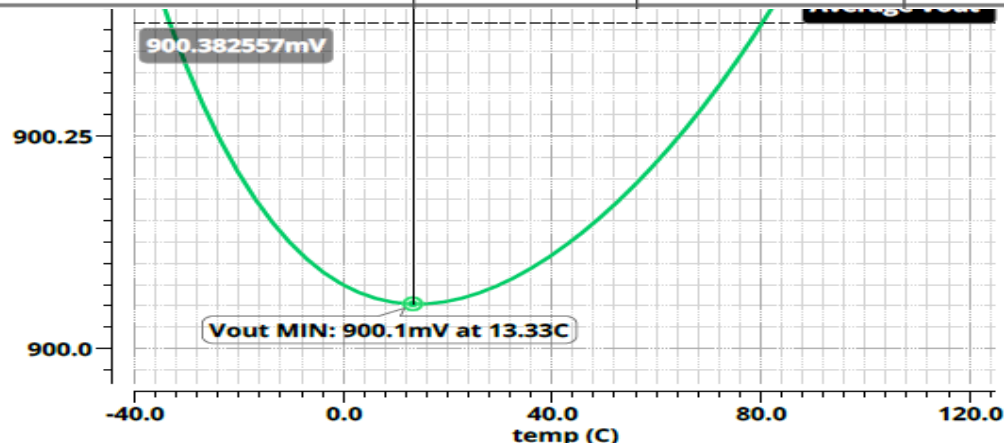
Temperature Coefficient - LDO (Pre LayOut)



Temperature Coefficient - LDO (Post LayOut)



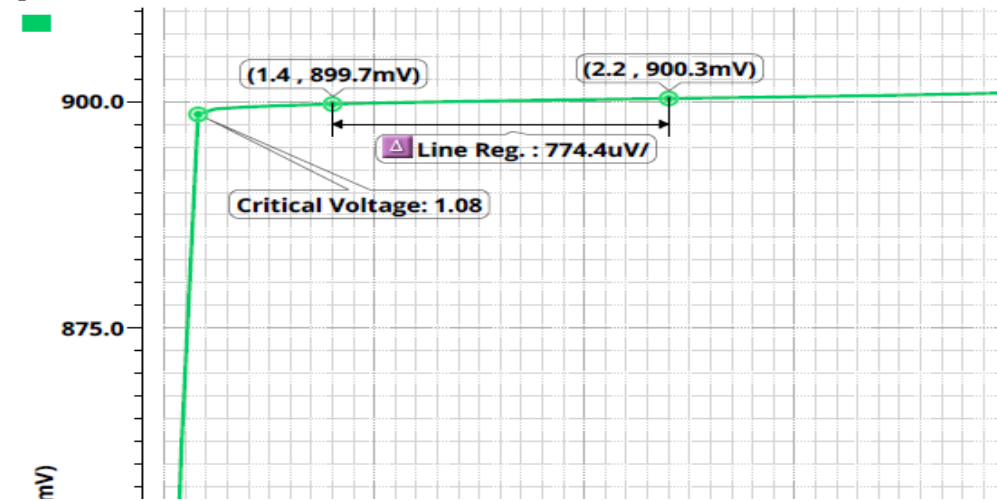
Parameter	Spec	Pre-LayOut	Post-LayOut	Range	Change (%)	Improvement from Spec
Temperature Coefficient (ppm/°C)	< 60	5.1	6.4	-40-125 °C	+25.5 %	↑ 9.4×



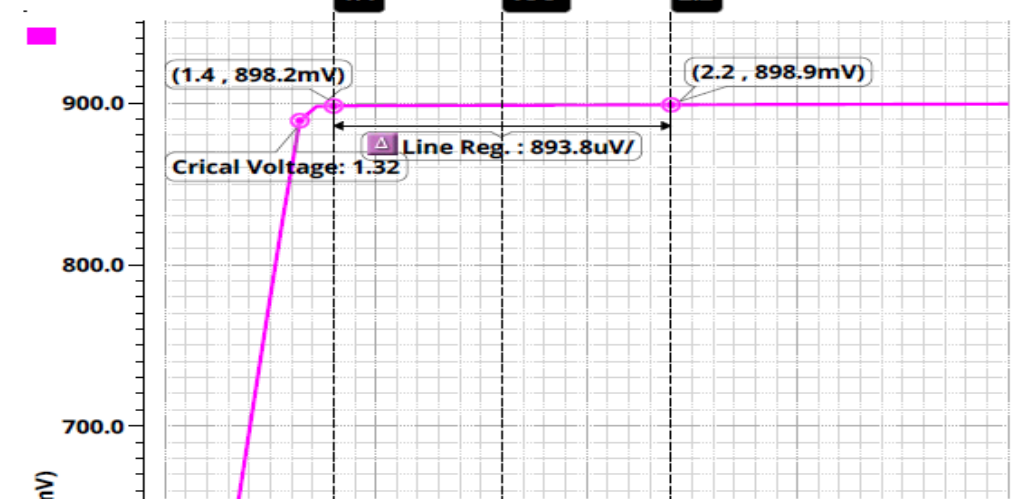
Temp Sweep : -40 – 125 [C], $I_{load} = 55[mA]$

LDO – Line Regulation

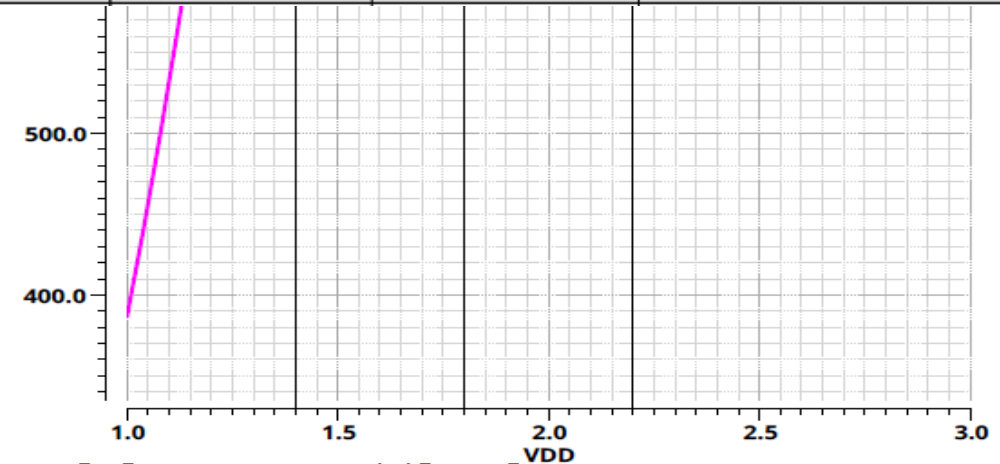
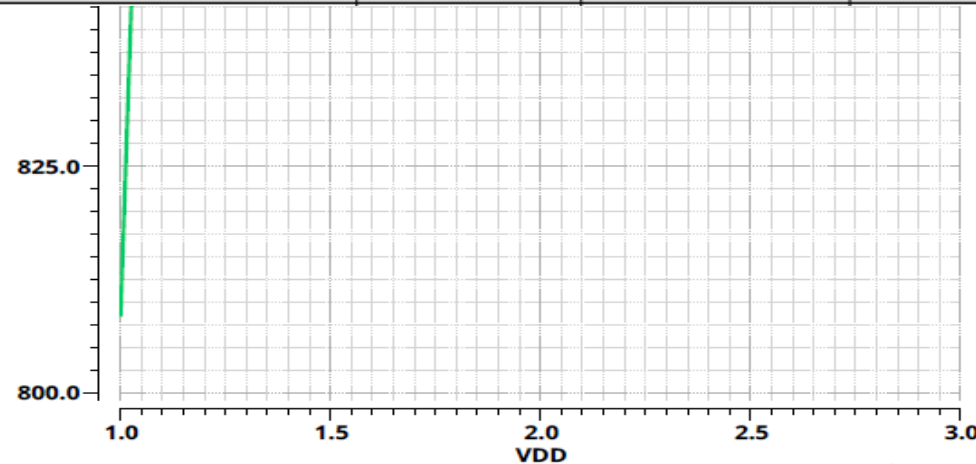
Line Regularization - LDO (Pre LayOut)



Line Regularization - LDO (Post LayOut)



Parameter	Spec	Pre-Layout	Post-Layout	Range	Change (%)	Improvement from Spec
Line Regulation (V/V)	< 0.01	0.0009	0.00089	1.4–2.2 V	–1.1 %	↑ 11.2×

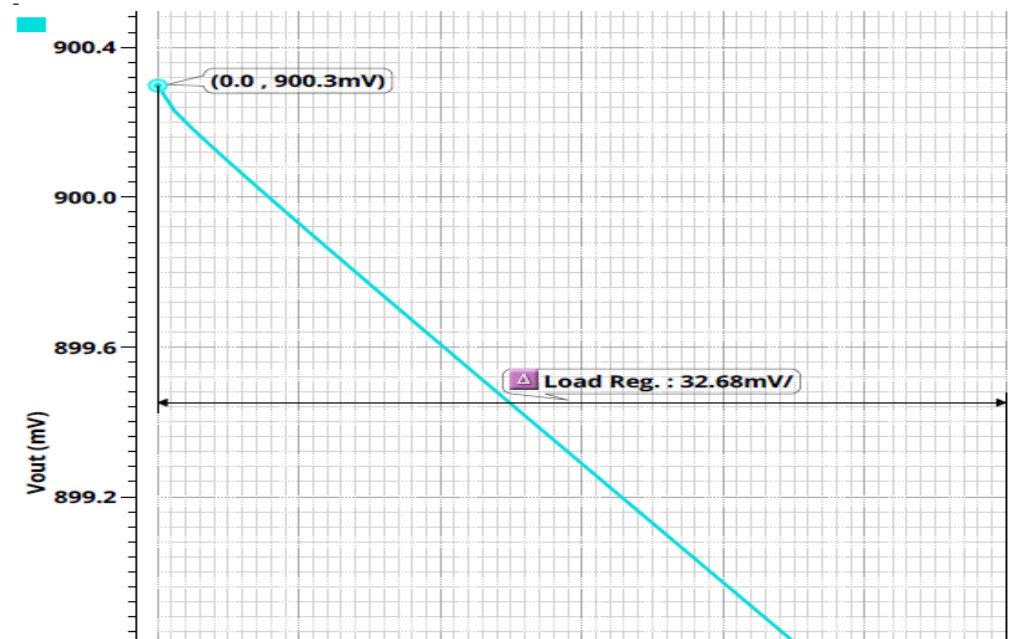


LDO – Load Regulation

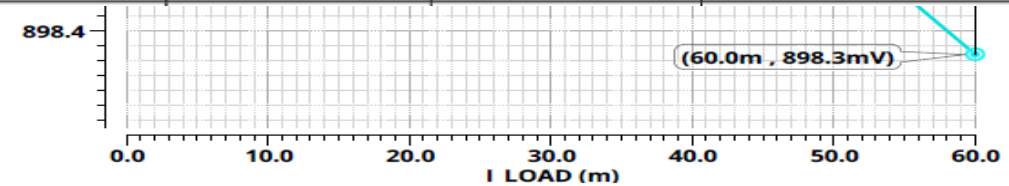
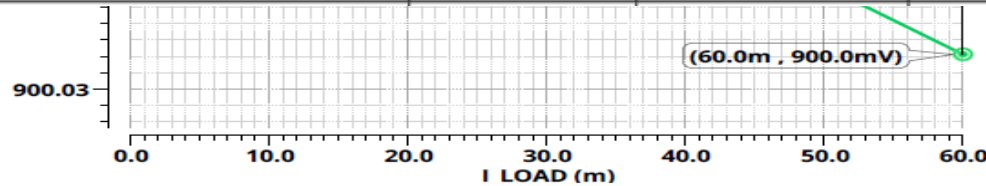
Load Regularization (Pre LayOut)



Load Regularization (Post LayOut)



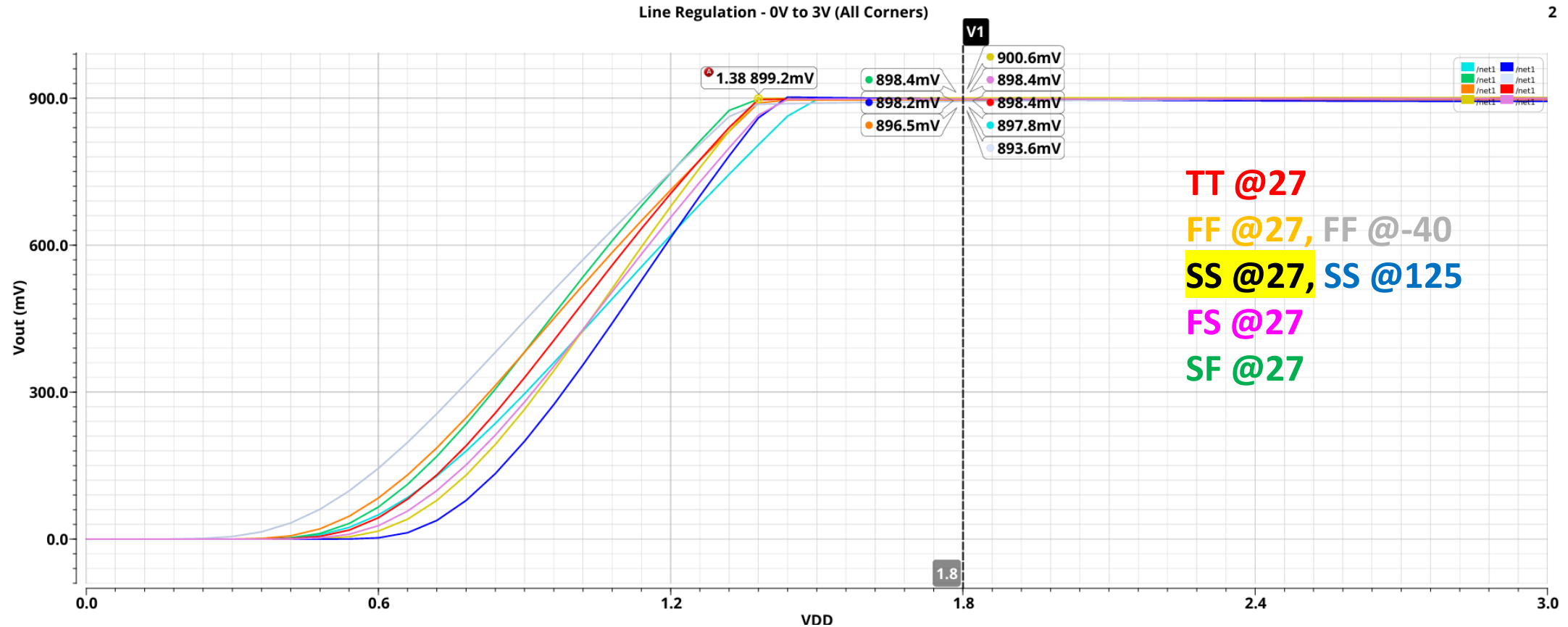
Parameter	Spec	Pre-Layout	Post-Layout	Range	Change (%)	Improvement from Spec
Load Regulation (mV/mA)	< 0.1	0.0024	0.033	0–60 mA	+1275 %	↑ 3×



$I_{load} \text{ Sweep} = 0 - 60[mA]$



LDO – Line Regulation across Corners

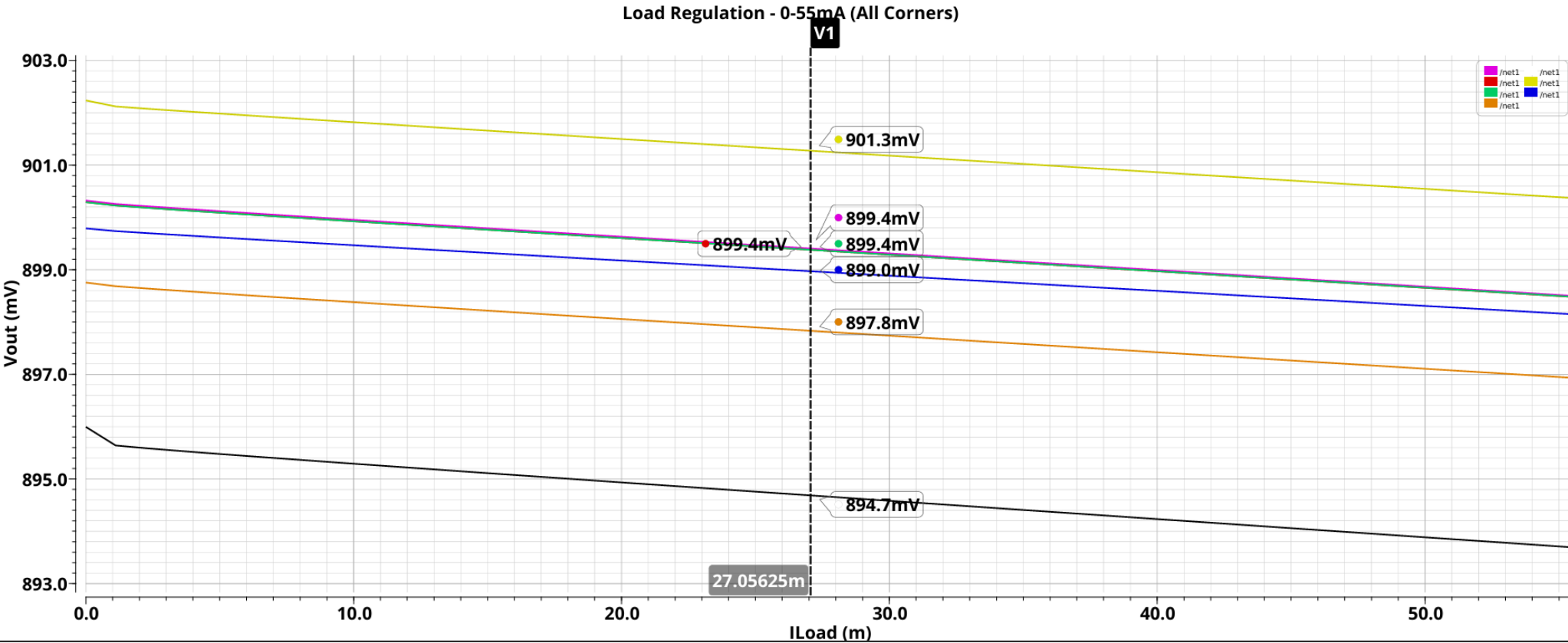


Parameter	Spec	Pre-Layout	Post-Layout	Range	Change (%)	Improvement from Spec
Line Regulation (V/V)	< 0.01	0.0009	0.00089	1.4–2.2 V	–1.1 %	↑ 11.2×



LDO – Load Regulation across Corners

3

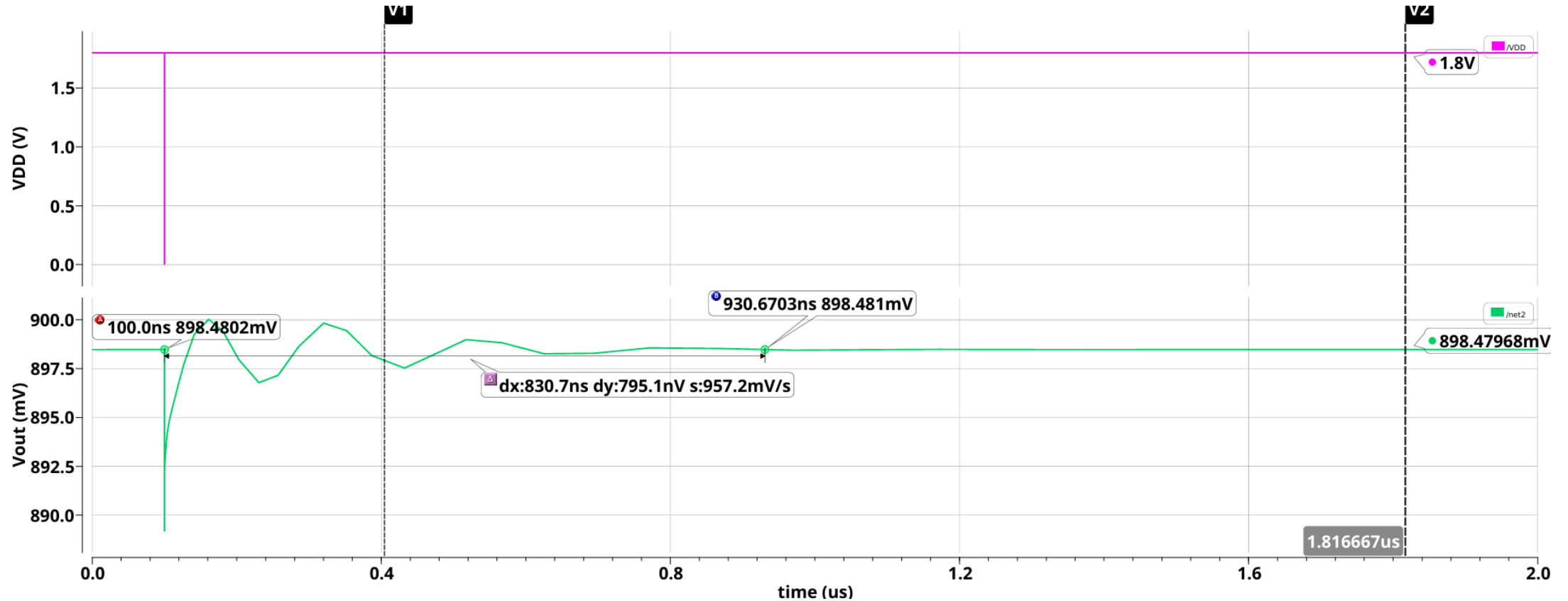


TT @27
FF @27, FF @-40
SS @27, SS @125
FS @27
SF @27

Parameter	Spec	Pre-Layout	Post-Layout	Range	Change (%)	Improvement from Spec
Load Regulation (mV/mA)	< 0.1	0.0024	0.033	0-60 mA	+1275 %	↑ 3×



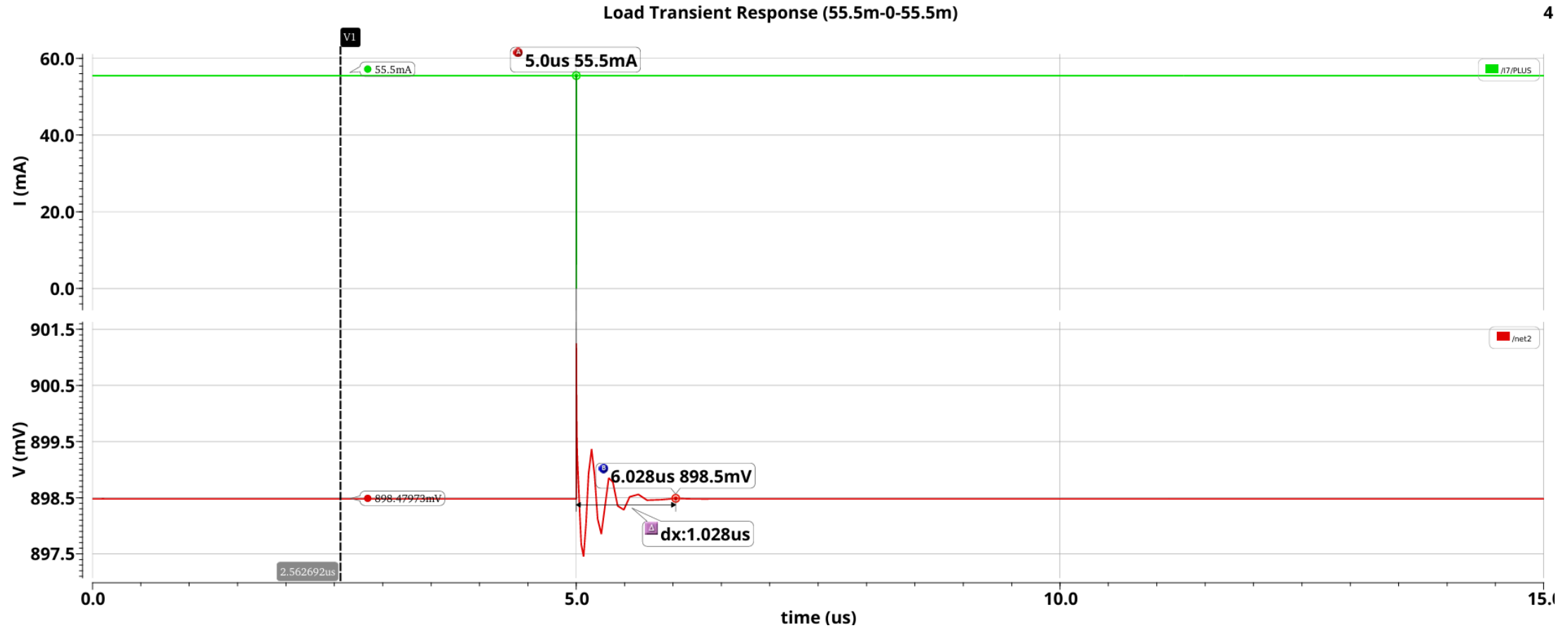
LDO – Line Transient to 10ps Disturbance



Phase Margin	GBW	Settling Time
57.81°	3.731MHz	0.8μS



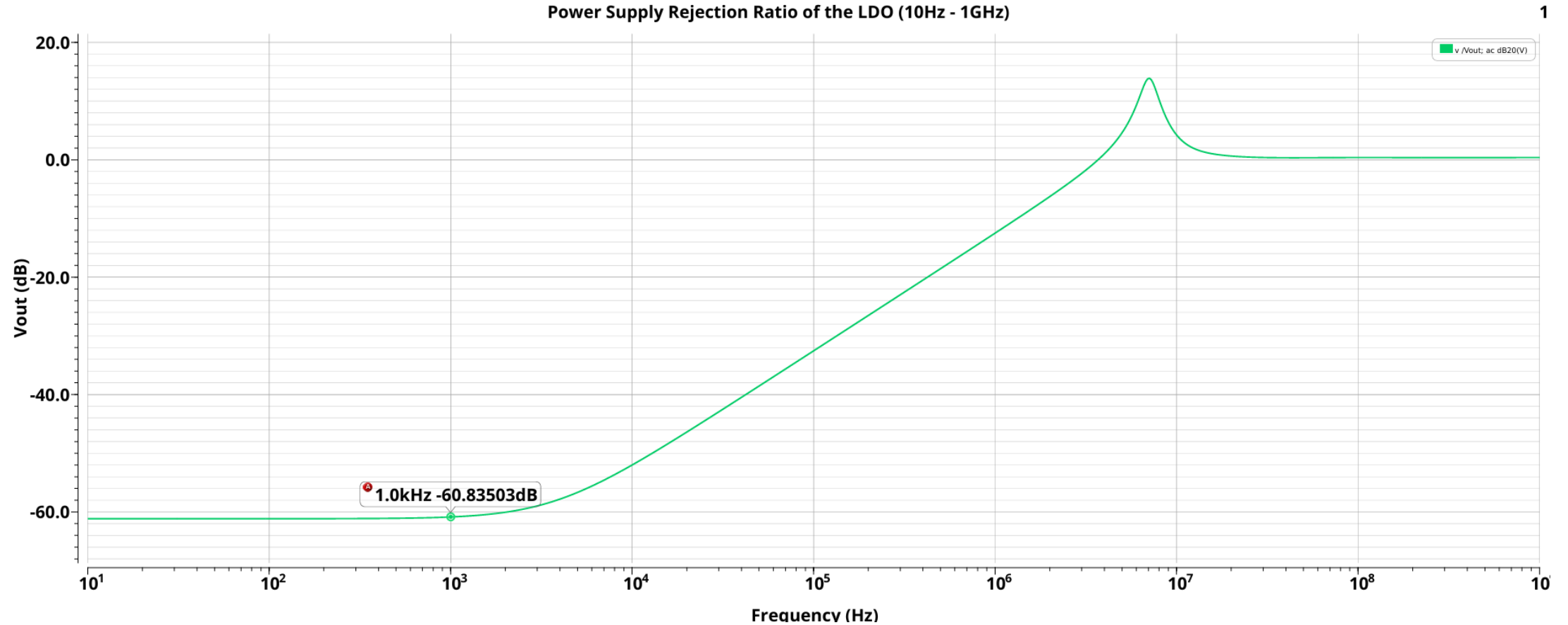
LDO – Load Transient to 10ps Disturbance



Phase Margin	GBW	Settling Time
57.81°	3.731MHz	0.8μs



LDO – Power Supply Rejection Ratio



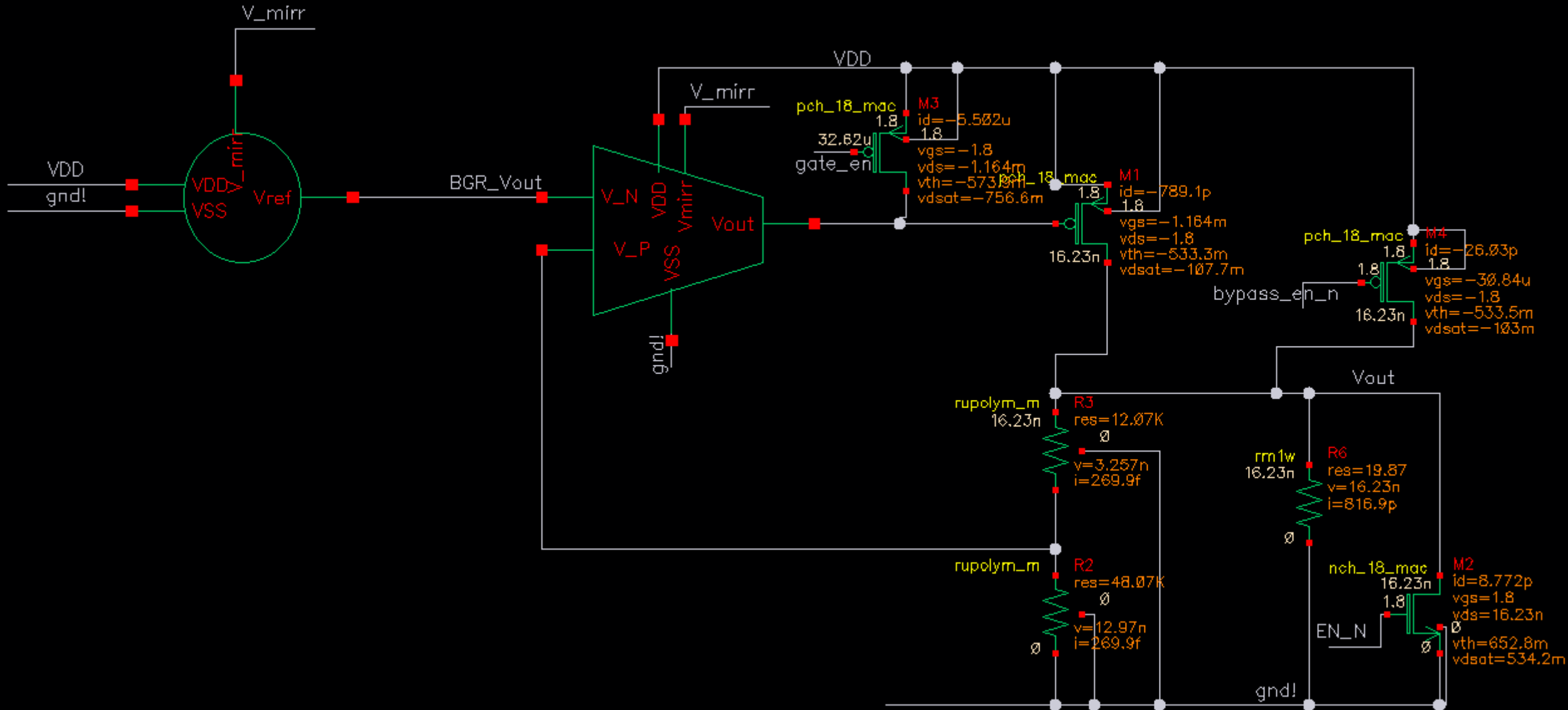
Parameter	Spec	Pre-Layout	Post-Layout	Range	Change (%)	Improvement from Spec
PSRR @ 1kHz	≤ -40 dB	-61 dB	-60.8 dB	N/A	-0.4 %	$\uparrow 11\times$



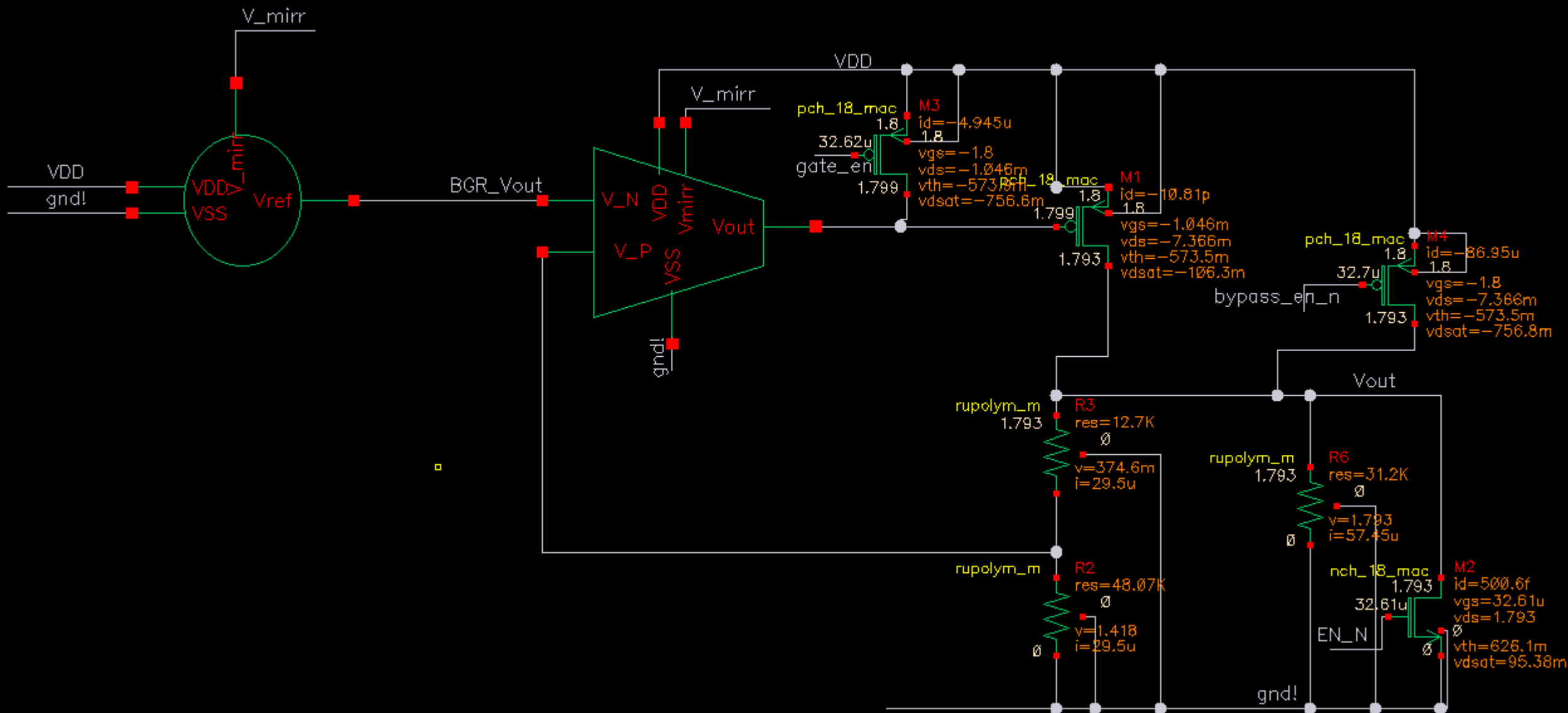
Full Post-Layout Results

Parameter	This Work
Process	28nm
Layout Area (mm ²)	0.0044
Supply Voltage V_{DD} (V)	1.38–3
Output Voltage V_{OUT} (V)	0.9
Load Current I_{Load} (mA)	55.5
Quiescent Current I_Q (μ A)	80
Line Regulation (mV/V)	0.89
Load Regulation (mV/mA)	0.033
Temperature Coefficient (ppm/ $^{\circ}$ C)	6.4
PSRR (dB @1kHz)	-60.8
Output Noise (nV/ $\sqrt{Hz}$ @100kHz)	672
Settling Time - Worst Case (μ s)	0.7

LDO – Disabled Mode



LDO – Bypass Mode





Conclusions

- ❑ A 0.9V LDO regulator was successfully implemented in 28nm CMOS for integration within a DPLL system-on-chip.
- ❑ The design met key specifications including line/load regulation, temperature stability, PSRR, transient response, area and, power constraints, demonstrating reliable performance across all PVT corners and scenarios.
- ❑ Major design challenges, such as balancing stability, gain, bandwidth, area, and power consumption tradeoffs, along with layout optimizations to improve matching and minimize IR drop, were effectively addressed through iterative refinement.
- ❑ Post-layout results showed only minor degradation, validating the effectiveness of early floorplanning and adherence to industry-standard layout practices.



Further Work

❑ EA Gain and BW Improvement:

- Use gain-boosted cascode or telescopic topologies (>100 dB gain) to enhance regulation and PSRR.
- **Tradeoffs:** Increased complexity, reduced output swing, and higher power.

❑ BGR Performance Enhancement:

- Implement second-order curvature correction and cascode current mirrors for sub-1 ppm/ $^{\circ}\text{C}$ temperature coefficient and improved PSRR.
- **Tradeoffs:** Higher power and design complexity.

❑ Pass Device Optimization:

- Widen pass transistor for better load regulation.
- **Tradeoffs:** Raises gate capacitance, reducing bandwidth and stability. Can be mitigated with a buffer stage.

❑ Advanced LDO Control Features:

- Add low-power modes and digitally tunable output voltage through feedback resistor ladder.
- **Tradeoffs:** More area and logic, requires early top-level integration.

❑ Output Noise Reduction:

- Employ low-noise OTA topologies or RC filtering on the BGR output to reduce low-frequency noise.
- **Tradeoffs:** Large passive RC filters impractical on chip. Other methods add complexity.

Figures

Description	Image Preview
LDO Block Diagram	Controls the output voltage by modulating current flow. 1.8[V] V_{DD} 0.9[V] V_{OUT} V_{ref} Error Amp V_g V_{fb} BGR OTA (Error Amplifier): adjusts the gate voltage of the PMOS pass device (i.e. V_{GS}) to deliver the required current Bandgap Reference: Provides a temperature-independent reference voltage. Feedback Network: Samples the output voltage for comparison with V_{ref}. $R1$ $R2$ Load I_{Load} $V_{fb} > V_{ref}$: $V_g \uparrow \Rightarrow V_{sg} \downarrow \Rightarrow I_{SD} \downarrow \Rightarrow V_{fb} \downarrow$ $V_{fb} < V_{ref}$: $V_g \downarrow \Rightarrow V_{sg} \uparrow \Rightarrow I_{SD} \uparrow \Rightarrow V_{fb} \uparrow$
Layout Screenshot	LDO OTA BGR OTA

DPLL-LDO